

POWERFACTORY

PowerFactory 2024

Technical Reference

Overhead Line Models

ElmLne, TypLne, TypGeo, TypTow

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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Contents

1 General Description **1**

1.1 Mutual Coupling 1

1.1.1 AC/DC Couplings 1

1.2 Frequency Dependency 2

1.3 Lumped and Distributed Parameters 2

1.4 Line Shunt Compensation 3

1.4.1 Input Mode 3

2 Equivalent Circuits **5**

2.1 Lumped Parameters 5

2.1.1 Single-Phase Line 6

2.1.2 Single-Phase Line with Neutral Conductor 7

2.1.3 Two-Phase Line 8

2.1.4 Two-Phase Line with Neutral Conductor 9

2.1.5 Three-Phase Lines 9

2.1.6 Three-Phase Line with Neutral Conductor 11

2.1.7 4-Wire Model Data Conversion 11

2.2 Distributed Parameter Model 15

2.2.1 General Formulation 16

3 Load Flow Analysis **19**

3.1 Temperature Dependency of Line Resistance 19

3.2 AC Model Calculation Parameters 19

3.3 AC Model Calculation Parameters for Linear DC Load Flow 22

3.4 DC Model Calculation Parameters 23

3.5 Consideration of Line Shunt Compensation 24

4 Short-Circuit **25**

4.1 Zero sequence mutual impedance 25

4.2 Consideration of Line Shunt Compensation 27

5 Harmonics Analysis **28**

5.1 Zero sequence mutual impedance 28

5.2 Consideration of Line Shunt Compensation 28

6 RMS Simulation **29**

6.1 Zero sequence mutual impedance 29

6.2 Consideration of Line Shunt Compensation 29

7 EMT Simulation **30**

7.1 Lumped Parameter Model 30

7.2 Distributed Parameter Model 31

7.2.1 Bergeron's Method for Solution in the Time Domain 31

7.2.2 Constant Parameter Model 33

7.2.3 Frequency Dependent Parameter (Modal) Model 35

7.2.4 Frequency Dependent Phase-Domain Models 37

7.3 Diagonalisation 41

7.4 Consideration of Line Shunt Compensation 41

A Parameter Definitions **43**

B References **46**

List of Figures **47**

List of Tables **48**

1 General Description

This document describes the models of transmission lines available in *PowerFactory*.

*PowerFactory*provides models which can represent dc and ac lines for all possible phase technologies (3-phase, 2-phase and 1-phase; with/without a neutral conductor and ground wires) for both single- and mutually-coupled parallel circuits. Table 1.1 provides an overview of the supported options and the corresponding element/type combinations.

Table 1.1: Overview of line models available in *PowerFactory*

System	Phase Technology	Element	Type
DC	Unipolar	*ElmLne*	*TypLne*
AC, single-circuit	1-ph	*ElmLne*	*TypLne*
	2-ph	*ElmLne*	*TypLne*
	3-ph	*ElmLne*	*TypLne*, *TypTow*, *TypGeo*
	1-ph with neutral	*ElmLne*	*TypLne*
	2-ph with neutral	*ElmLne*	*TypLne*
	3-ph with neutral	*ElmLne*	*TypLne*
AC, mutually-coupled circuits	Any combination of phase technologies	*ElmTow*	*TypTow*, *TypGeo*
DC, mutually-coupled circuits	Any combination of phase technologies	*ElmTow*	*TypTow*, *TypGeo*
AC/DC, mutually-coupled circuits	Any combination of phase technologies	*ElmTow*	*TypTow*

The line element *ElmLne* in *PowerFactory* is the constituent element of transmission lines. When referring to a type, the line element can be used to define single-circuit lines of any phase technology according to Table 1.1. In addition, the element parameter *Number of Parallel Lines* allows the representation of parallel lines without mutual coupling.

1.1 Mutual Coupling

If the mutual coupling between parallel lines is to be considered, then a line coupling element *ElmTow* has to be defined. In this case, the line element *ElmLne* points to a line coupling element *ElmTow* which in turns refers to the corresponding tower type (*TypTow*) or tower geometry type (*TypGeo*).

Alternatively zero sequence mutual impedance objects (*ElmZ0mut*) can be used to define zero sequence coupling impedances of three phase lumped parameter line models. For details, see section 4.1.

1.1.1 AC/DC Couplings

AC and DC circuits increasingly share rights-of-way, often with only small distances between the transmission lines, resulting in electromagnetic coupling effects. Voltage and current can be induced on either AC or DC lines under both steady-state and transient conditions. AC/DC tower couplings can be modelled by using the line coupling element (*ElmTow*) in conjunction with the tower type (*TypTow*). The tower type must have the *System Type* set to *AC/DC*. The impedances and admittances of the AC circuit/s will be solved using the user-defined *Nominal*

Frequency, and of the DC circuit/s using the *Nominal Frequency (DC)*. AC/DC couplings are handled as follows in the various *PowerFactory* calculations:

- Load Flow: The transmission line equations are solved separately for the AC circuit/s and DC circuit/s. The DC circuit/s are always solved using the lumped parameter model, regardless of the selection of the *Line Model* in the line coupling element (*ElmTow*).
- Short-Circuit: AC circuit/s are considered and any DC circuit/s are ignored.
- RMS Simulation: The DC circuit/s are always solved using the lumped parameter model, regardless of the selection of the *Line Model* in the line coupling element (*ElmTow*).
- EMT Simulation: The frequency dependent phase-domain model is selected by default in the line coupling element (*ElmTow*). The user can choose between the Universal Line Model (ULM) or the Frequency Dependent Model (FDM).

1.2 Frequency Dependency

PowerFactory further distinguishes between constant and frequency dependent parameter models. Models based on tower geometry types (*TypTow* or *TypGeo*) use frequency dependent parameters. This means that the electrical parameters of the line per unit length are calculated from the mechanical characteristics of the tower and the conductors accounting for skin effect, the frequency dependent earth-return path of the line, etc. These types should be selected for use in simulations where a wide range of frequencies is involved or frequencies other than the nominal system frequency. For further information about the calculation of the per unit length parameters, please refer to [1].

Models based on line types (*TypLne*) are by default not frequency dependent. The user enters the electrical parameters per unit length of the line at system frequency. These parameters remain unchanged; if the frequency of the simulation changes, i.e. differs from the system frequency, then the program will adjust the reactance and susceptance of the line according to the new frequency but the inductances and capacitances will remain unchanged. For certain calculations (i.e. harmonic load flow, frequency sweep) the user still has the option of assigning a frequency characteristic to the parameters in the line type. Further details pertaining to input parameters for the different phase technologies (3-phase, 2-phase, 1-phase, w/o neutral) and frequency characteristics are discussed in the following sections.

1.3 Lumped and Distributed Parameters

For three-phase lines (either single or multiple parallel circuits), the user can choose between lumped or distributed parameters. For long transmission lines the distributed parameter model is preferred as it gives highly accurate results, while the lumped parameter model provides sufficient results for short lines. The details of the different models are discussed in the following sections.

Rating of the line/cable

The rated current is the base for per-unit current quantities. The rated current I_r of the three-phase line/cable is calculated as:

$$I_r = sline \cdot nlnum \quad \mathrm{kA} \tag{1}$$

or for a cable with a selected line type (*TypLne*) and the option *Air* for laying is selected:

$$I_r = InomAir \cdot nlnum \quad \mathrm{kA} \tag{2}$$

where:

- $sline$ is the rated current in kA from the line type (*TypLne*, *sline*). In case of using a tower type or a tower geometry the rated current (*sline*) from the corresponding conductor type (*TypCon*) is used. Note the rated current of the conductor type is the total rated current of the whole bundle.
- $nlnum$ is the number of parallel lines from
- $InomAir$ is the rated current in air in kA from the line type (*TypLne*)

If line sections are defined in the line/cable, the smallest rated current of all sections is used as the rated current.

1.4 Line Shunt Compensation

Shunt reactors are used to compensate reactive power generation in long transmission lines, thereby increasing the efficiency and power quality of the network. Transmission line elements (*ElmLne*) in *PowerFactory* can be defined to have shunt compensation at each end of the line via the use of Line Compensation elements (*ElmLnecomp*). Such elements are accessible either via the *Sections/Line Loads/Compensation* button on the *Basic Data* page or via the *Compensation* subpage. In addition, the Project Settings (*Calculation Options* $\rightarrow$ *Lines* page) provide an option to consider the line compensation current when calculating the line loading.

1.4.1 Input Mode

The two input modes and associated input parameters available in the Line Compensation element (*ElmLnecomp*) are described below.

1.4.1.1 Power values

Input fields are available for $P1$ and $P0$ in MW, and $Q1$ and $Q0$ in Mvar. The *Compensation type* is automatically adjusted to *R-L* or *C* based on the sign of $Q1$, as illustrated in Figure 1.1.

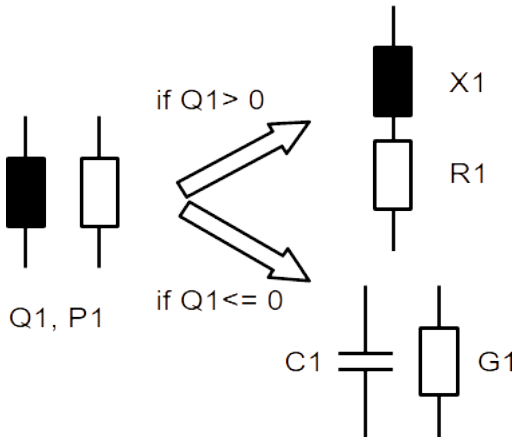


Figure 1.1: Positive sequence input parameters

1.4.1.2 Design parameters

Input fields are available for the *Reactive power* Q_{rea} and *Quality factor at fn* $qfrea$, *R0/R1 ratio* $R0/R1$ and *X0/X1 ratio*, $X0/X1$.

The conversion from *Reactive power*, Q_{rea}, and *Quality factor* to $P1$ and $Q1$ is performed according to:

$$\begin{aligned} P1 + j \cdot Q1 &= \frac{Q_{rea} \cdot (1/q_{frea} + j)}{(1/q_{frea})^2 + 1} \\ P1 &= \frac{Q_{rea} \cdot 1/q_{frea}}{(1/q_{frea})^2 + 1} \\ Q1 &= \frac{Q_{rea}}{(1/q_{frea})^2 + 1} \end{aligned} \tag{3}$$

and vice versa:

$$\begin{aligned} Q_{rea} &= Q1 \cdot (1/q_{frea})^2 + 1) \\ q_{frea} &= \frac{Q1}{P1} \end{aligned} \tag{4}$$

If $q_{frea} = 0$, then $P1 = 0$ and $Q1 = Q_{rea}$.

The frequency dependence of the impedance can be defined on the *Power Quality/Harmonics* page of the element using a relative characteristic.

2 Equivalent Circuits

2.1 Lumped Parameters

Figure 2.1 shows the equivalent PI-circuit used by *PowerFactory* to represent AC transmission lines with lumped parameters. The subscripts *s* and *r* denote the *sending* and *receiving* ends of the line, respectively. The general formulation discussed in this section is valid for any phase technology by appropriate dimensioning of the impedance and admittance matrices, even though the formulation presented here is based on a 3-phase line with no neutral conductor.

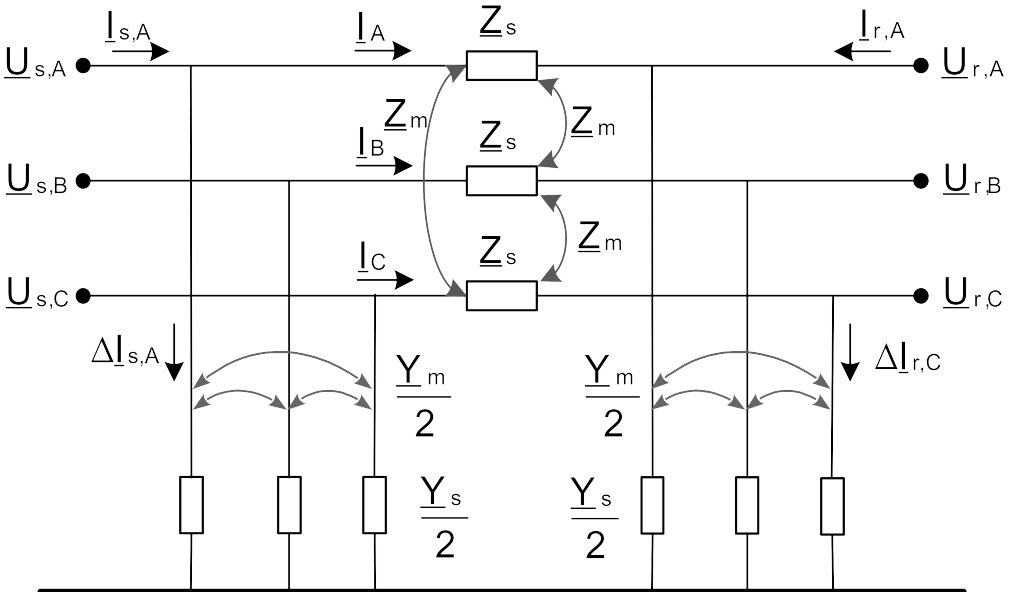


Figure 2.1: Equivalent PI-circuit of the line for lumped parameters

The equations of the voltages and currents at the sending and receiving ends of the line are formulated in terms of impedance and admittance matrices. The dimension of the matrices depends on the phase technology. The longitudinal voltage drop along the line is given by the impedance matrix in the following form:

$$\begin{bmatrix} \underline{U}_{s,A} \\ \underline{U}_{s,B} \\ \underline{U}_{s,C} \end{bmatrix} - \begin{bmatrix} \underline{U}_{r,A} \\ \underline{U}_{r,B} \\ \underline{U}_{r,C} \end{bmatrix} = \begin{bmatrix} \Delta\underline{U}_A \\ \Delta\underline{U}_B \\ \Delta\underline{U}_C \end{bmatrix} = \begin{bmatrix} \underline{Z}_s & \underline{Z}_m & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_s & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_m & \underline{Z}_s \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_A \\ \underline{I}_B \\ \underline{I}_C \end{bmatrix} \tag{5}$$

According to the sign convention assumed in Figure 2.1, the current at the sending end of the line is calculated in terms of the admittance matrix as follows:

$$\begin{bmatrix} \underline{I}_{s,A} \\ \underline{I}_{s,B} \\ \underline{I}_{s,C} \end{bmatrix} = \begin{bmatrix} \Delta\underline{I}_{s,A} \\ \Delta\underline{I}_{s,B} \\ \Delta\underline{I}_{s,C} \end{bmatrix} + \begin{bmatrix} \underline{I}_A \\ \underline{I}_B \\ \underline{I}_C \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \underline{Y}_s & \underline{Y}_m & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_s & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_m & \underline{Y}_s \end{bmatrix} \cdot \begin{bmatrix} \underline{U}_{s,A} \\ \underline{U}_{s,B} \\ \underline{U}_{s,C} \end{bmatrix} + \begin{bmatrix} \underline{I}_A \\ \underline{I}_B \\ \underline{I}_C \end{bmatrix} \tag{6}$$

Similarly, the current at the receiving end of the line is given by:

$$\begin{bmatrix} \underline{I}_{r,A} \\ \underline{I}_{r,B} \\ \underline{I}_{r,C} \end{bmatrix} = \begin{bmatrix} \Delta\underline{I}_{r,A} \\ \Delta\underline{I}_{r,B} \\ \Delta\underline{I}_{r,C} \end{bmatrix} - \begin{bmatrix} \underline{I}_A \\ \underline{I}_B \\ \underline{I}_C \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \underline{Y}_s & \underline{Y}_m & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_s & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_m & \underline{Y}_s \end{bmatrix} \cdot \begin{bmatrix} \underline{U}_{r,A} \\ \underline{U}_{r,B} \\ \underline{U}_{r,C} \end{bmatrix} - \begin{bmatrix} \underline{I}_A \\ \underline{I}_B \\ \underline{I}_C \end{bmatrix} \quad (7)$$

Equations (5), (6) and (7) completely define the PI-model of the line for lumped parameters. The impedance and admittance matrices:

$$[\underline{\boldsymbol{Z}}_{ABC}] = \begin{bmatrix} \underline{Z}_s & \underline{Z}_m & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_s & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_m & \underline{Z}_s \end{bmatrix} \qquad [\underline{\boldsymbol{Y}}_{ABC}] = \begin{bmatrix} \underline{Y}_s & \underline{Y}_m & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_s & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_m & \underline{Y}_s \end{bmatrix} \quad (8)$$

are the natural impedance and admittance matrices of the line after reduction of earth wires (if any).

Note that $\underline{Y}_s$ represents the sum of all admittances connected to the corresponding phase, while $\underline{Y}_m$ is the negative value of the admittance between two phases. Similarly, $\underline{Y}_p$ is the sum of all admittances connected to the neutral conductor and $\underline{Y}_{pn}$ is the negative value of the admittance between the neutral and the phase conductors.

The PI-circuit described here is the general formulation of the line model with lumped parameters in *PowerFactory*. The following sections discuss the application of the model to the different phase technologies (3-, 2-, 1-phase, w/o neutral conductors) and the required user-defined parameters in each case.

2.1.1 Single-Phase Line

The equivalent circuit in Figure 2.1 can be reduced to the PI-circuit in Figure 2.2 for the single-phase line.

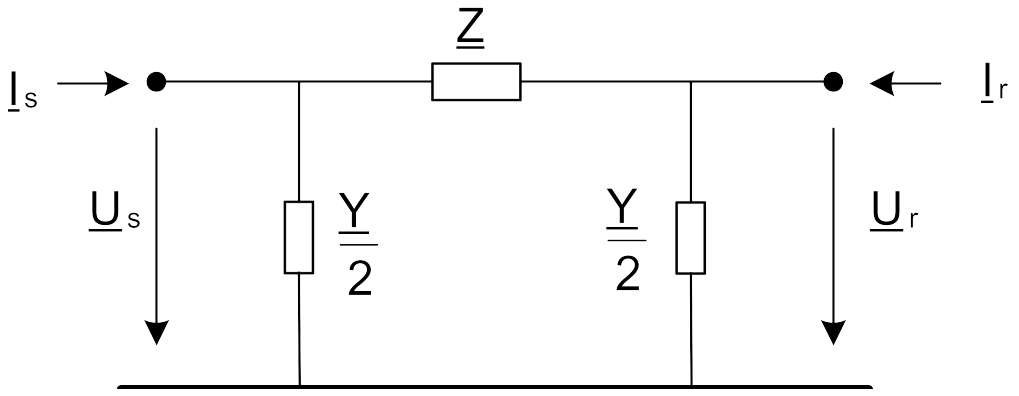


Figure 2.2: Lumped parameter model for a single-phase AC line

The impedance and admittance of the equivalent circuit are calculated from the input parameters defined in the line type (*TypLne*) according to the following equations:

$$\begin{aligned} Z &= Z_1^{'} \cdot l = \left(R_1^{'} + j\omega L_1^{'}\right) \cdot l \\ Y &= Y_1^{'} \cdot l = \left(G_1^{'} + j\omega C_1^{'}\right) \cdot l \\ G_1^{'} &= B_1^{'} \cdot tg\delta_1 \end{aligned} \quad (9)$$

where l is the length of the line in km, and $R_1^{'}$,$L_1^{'}$, $G_1^{'}$ and $C_1^{'}$ are the line parameters per unit length. Note that the conductance $G_1^{'}$ can be defined in terms of the insulation factor, $tg\delta$. The reader is referred to Table A.1 for the complete list of input parameters.

The currents and voltages on both sides of the line in Figure 2.2 are related by the following equation:

$$\begin{bmatrix} U_s \\ I_s \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \cdot \begin{bmatrix} U_r \\ -I_r \end{bmatrix} \tag{10}$$

where the *ABCD* parameters of the equivalent PI-circuit are:

$$\begin{aligned} A &= 1 + \tfrac{1}{2} Z^{'} \cdot Y^{'} \cdot l^2 \\ B &= Z^{'} \cdot l \\ C &= Y^{'} \cdot l \cdot \left(1 + \tfrac{Z^{'} \cdot Y^{'} \cdot l^2}{4}\right) \\ D &= A \end{aligned} \tag{11}$$

2.1.2 Single-Phase Line with Neutral Conductor

From Figure 2.1, the equivalent circuit shown in Figure 2.3 can be now deduced for the single-phase line with neutral conductor model.

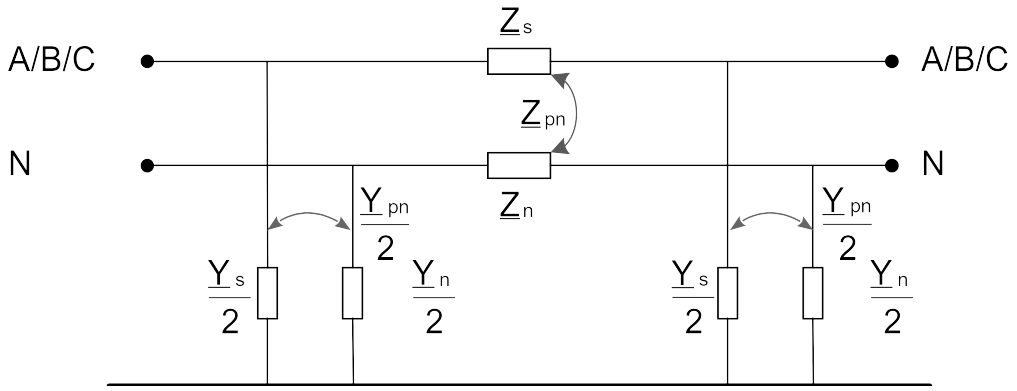


Figure 2.3: Equivalent circuit of the single-phase line with neutral conductor model

The self- and mutual impedances and admittances of the equivalent circuit:

$$[\underline{\boldsymbol{Z}}_{an}] = \begin{bmatrix} \underline{Z}_s & \underline{Z}_{pn} \\ \underline{Z}_{pn} & \underline{Z}_n \end{bmatrix} \qquad [\underline{\boldsymbol{Y}}_{an}] = \begin{bmatrix} \underline{Y}_s & \underline{Y}_{pn} \\ \underline{Y}_{pn} & \underline{Y}_s \end{bmatrix} \tag{12}$$

are calculated from the input parameters defined in the line type (*TypLne*).

Thus the self and mutual impedances and admittances in (12) are related to the input parameters $\underline{Z}_1$, $\underline{Z}_n$, $\underline{Z}_{pn}$, $\underline{Y}_1$, $\underline{Y}_n$ and $\underline{Y}_{pn}$ as follows:

$$\underline{Z_s} = \underline{Z}_1 \qquad \underline{Y_s} = \underline{Y}_1 \tag{13}$$

The self- and mutual impedance $\underline{Z}_n$, $\underline{Z}_{pn}$ and the admittance for the neutral conductor $\underline{Y}_n$, $\underline{Y}_{pn}$ are as listed in Table A.1 (Input parameter of the line type (*TypLne*)) of annex A.

2.1.3 Two-Phase Line

From Figure 2.1, the equivalent circuit shown in Figure 2.4 can be now deduced for the 2-phase line model.

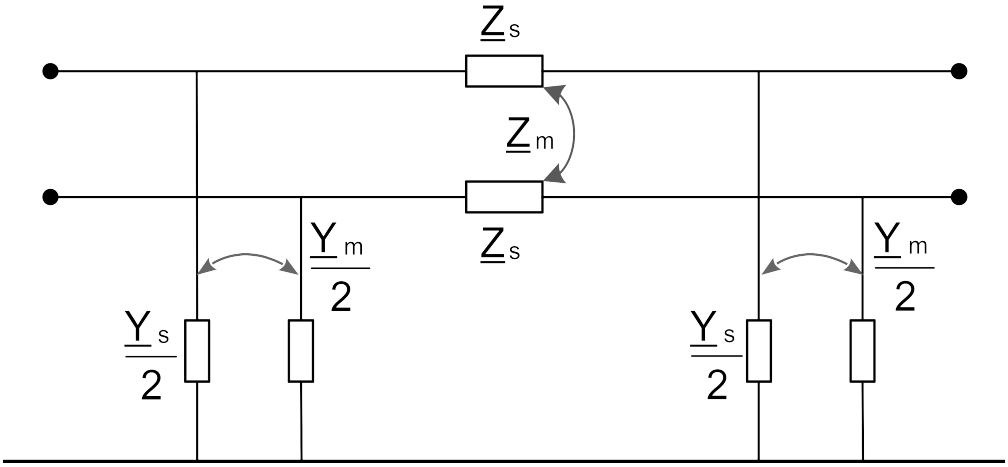


Figure 2.4: Equivalent circuit of the two-phase line model

The self- and mutual impedances and admittances of the equivalent circuit:

$$[\underline{\boldsymbol{Z}}_{ab}] = \begin{bmatrix} \underline{Z}_s & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_s \end{bmatrix} \qquad [\underline{\boldsymbol{Y}}_{ab}] = \begin{bmatrix} \underline{Y}_s & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_s \end{bmatrix} \tag{14}$$

are calculated from the input parameters defined in the line type (*TypLne*). The input parameters - positive and zero sequence components - are converted to the impedances and admittances in (14) via the following transformation:

$$[\underline{\boldsymbol{T}}_{S2ph}] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad [\underline{\boldsymbol{T}}_{S2ph}]^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$[\underline{\boldsymbol{Z}}_{01}] = [\boldsymbol{T}_{S2ph}]^{-1} \times [\underline{\boldsymbol{Z}}_{ab}] \times [\boldsymbol{T}_{S2ph}]$$

Thus the self and mutual impedances and admittances in (14) are related to the input parameters $\underline{Z}_1$, $\underline{Z}_0$, $\underline{Y}_1$ and $\underline{Y}_0$ as follows:

$$\underline{Z}_s = \frac{1}{2} \cdot (\underline{Z}_0 + \underline{Z}_1) \qquad \underline{Y}_s = \frac{1}{2} \cdot (\underline{Y}_0 + \underline{Y}_1) \tag{15}$$

$$\underline{Z}_m = \frac{1}{2} \cdot (\underline{Z}_0 - \underline{Z}_1) \qquad \underline{Y}_m = \frac{1}{2} \cdot (\underline{Y}_0 - \underline{Y}_1) \tag{16}$$

or can be resolved in terms of the sequence magnitudes:

$$[\underline{\boldsymbol{Z}}_{01}] = \begin{bmatrix} \underline{Z}_0 & 0 \\ 0 & \underline{Z}_1 \end{bmatrix} = \begin{bmatrix} \underline{Z}_s + \underline{Z}_m & 0 \\ 0 & \underline{Z}_s - \underline{Z}_m \end{bmatrix}$$

$$[\underline{\boldsymbol{Y}}_{01}] = \begin{bmatrix} \underline{Y}_0 & 0 \\ 0 & \underline{Y}_1 \end{bmatrix} = \begin{bmatrix} \underline{Y}_s + \underline{Y}_m & 0 \\ 0 & \underline{Y}_s - \underline{Y}_m \end{bmatrix}$$

2.1.4 Two-Phase Line with Neutral Conductor

From Figure 2.1, the equivalent circuit shown in Figure 2.5 can be now deduced for the 2-phase line with neutral conductor model.

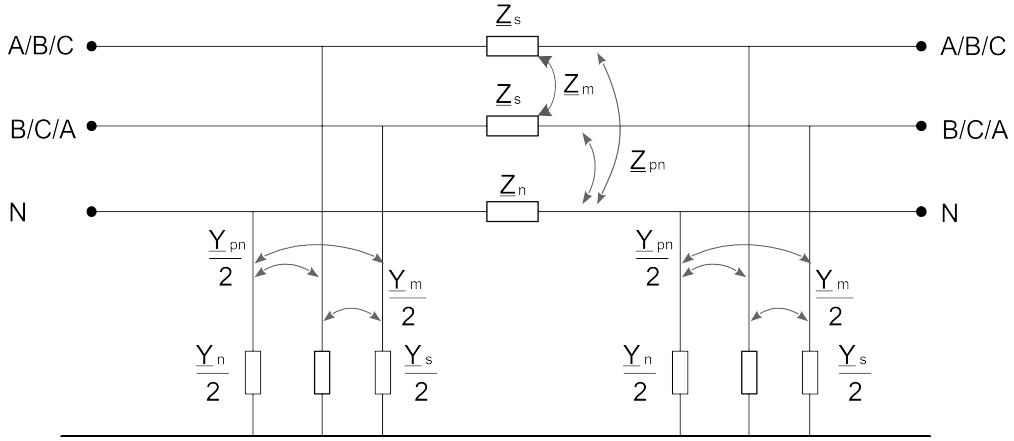


Figure 2.5: Equivalent circuit of the two-phase line with neutral conductor model

The self- and mutual impedances and admittances of the equivalent circuit are:

$$[\underline{\boldsymbol{Z}}_{abn}] = \begin{bmatrix} \underline{Z}_s & \underline{Z}_m & \underline{Z}_{pn} \\ \underline{Z}_m & \underline{Z}_s & \underline{Z}_{pn} \\ \underline{Z}_{pn} & \underline{Z}_{pn} & \underline{Z}_n \end{bmatrix} \quad [\underline{\boldsymbol{Y}}_{abn}] = \begin{bmatrix} \underline{Y}_s & \underline{Y}_m & \underline{Y}_{pn} \\ \underline{Y}_m & \underline{Y}_s & \underline{Y}_{pn} \\ \underline{Y}_{pn} & \underline{Y}_{pn} & \underline{Y}_n \end{bmatrix} \tag{17}$$

The input parameters of the model are the positive and zero sequence impedances $\underline{Z}_1$, $\underline{Z}_0$. The positive and zero sequence admittance, $\underline{Y}_1$ and $\underline{Y}_0$, the self- and mutual impedance $\underline{Z}_n$, $\underline{Z}_{pn}$ and the admittance for the neutral conductor $\underline{Y}_n$, $\underline{Y}_{pn}$ are as listed in Table A.1 (Input parameter of the line type (*TypLne*)) of annex A.

The values $\underline{Z}_n$, $\underline{Z}_{pn}$, $\underline{Y}_p$ and $\underline{Y}_{pn}$ of the neutral conductor can be directly used in (17).

Thus the self and mutual impedances and admittances in (17) are related to the input parameters $\underline{Z}_1$, $\underline{Z}_0$, $\underline{Y}_1$ and $\underline{Y}_0$ and are determined like for the two phase line model (see equations (15) and (16)).

2.1.5 Three-Phase Lines

The equivalent circuit of the three-phase line is shown in Figure 2.6. The self- and mutual impedances and admittances are given by:

$$[\underline{\boldsymbol{Z}}_{abc}] = \begin{bmatrix} \underline{Z}_s & \underline{Z}_m & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_s & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_m & \underline{Z}_s \end{bmatrix} \qquad [\underline{\boldsymbol{Y}}_{abc}] = \begin{bmatrix} \underline{Y}_s & \underline{Y}_m & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_s & \underline{Y}_m \\ \underline{Y}_m & \underline{Y}_m & \underline{Y}_s \end{bmatrix} \tag{18}$$

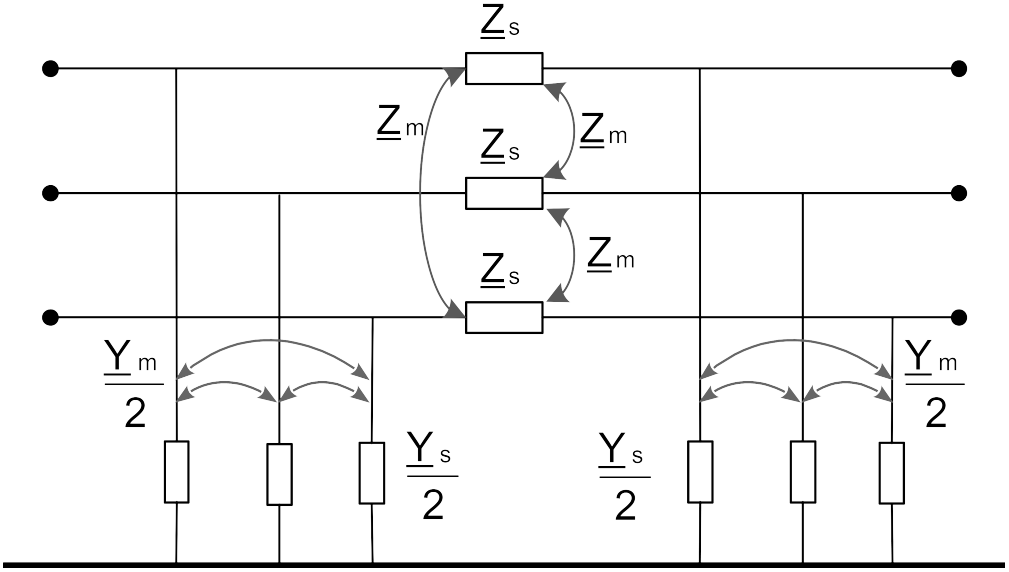


Figure 2.6: Equivalent circuit of the three-phase line

The input parameters in the line type (*TypLne*) are defined in terms of positive and zero sequence impedances and admittances $\underline{Z}_1, \underline{Y}_1, \underline{Z}_0$ and $\underline{Y}_0$. The negative sequence is assumed to be equal to the positive sequence.

The conversion from the sequence components into the natural components in (18) is done via the complex transformation matrix $[T_s]$ as follows:

$$[\underline{\boldsymbol{T}}_S] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \quad \rightarrow \quad [\underline{\boldsymbol{T}}_S]^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \tag{19}$$

$$[\underline{\boldsymbol{Z}}_{012}] = [\boldsymbol{T}_S]^{-1} \times [\underline{\boldsymbol{Z}}_{abc}] \times [\boldsymbol{T}_S]$$

Thus the self and mutual impedances and admittances in (18) are related to the input parameters $\underline{Z}_1$, $\underline{Z}_0$, $\underline{Y}_1$ and $\underline{Y}_0$ as follows:

$$\underline{Z}_s = \frac{1}{3} \cdot (\underline{Z}_0 + 2 \cdot \underline{Z}_1) \qquad \underline{Y}_s = \frac{1}{3} \cdot (\underline{Y}_0 + 2 \cdot \underline{Y}_1) \tag{20}$$

$$\underline{Z}_m = \frac{1}{3} \cdot (\underline{Z}_0 - \underline{Z}_1) \qquad \underline{Y}_m = \frac{1}{3} \cdot (\underline{Y}_0 - \underline{Y}_1) \tag{21}$$

or can be resolved in terms of the sequence magnitudes:

$$[\underline{\boldsymbol{Z}}_{012}] = \begin{bmatrix} \underline{Z}_0 & 0 & 0 \\ 0 & \underline{Z}_1 & 0 \\ 0 & 0 & \underline{Z}_2 \end{bmatrix} = \begin{bmatrix} \underline{Z}_s + 2\underline{Z}_m & 0 & 0 \\ 0 & \underline{Z}_s - \underline{Z}_m & 0 \\ 0 & 0 & \underline{Z}_s - \underline{Z}_m \end{bmatrix}$$

$$[\underline{\boldsymbol{Y}}_{012}] = \begin{bmatrix} \underline{Y}_0 & 0 & 0 \\ 0 & \underline{Y}_1 & 0 \\ 0 & 0 & \underline{Y}_2 \end{bmatrix} = \begin{bmatrix} \underline{Y}_s + 2\underline{Y}_m & 0 & 0 \\ 0 & \underline{Y}_s - \underline{Y}_m & 0 \\ 0 & 0 & \underline{Y}_s - \underline{Y}_m \end{bmatrix}$$

2.1.6 Three-Phase Line with Neutral Conductor

Figure 2.7 shows the equivalent circuit of the 3-phase line with neutral conductor. The voltages and the currents at both ends of the line are related by the impedance and admittance matrices:

$$[\underline{\boldsymbol{Z}}_{abcn}] = \begin{bmatrix} \underline{Z}_s & \underline{Z}_m & \underline{Z}_m & \underline{Z}_{pn} \\ \underline{Z}_m & \underline{Z}_s & \underline{Z}_m & \underline{Z}_{pn} \\ \underline{Z}_m & \underline{Z}_m & \underline{Z}_s & \underline{Z}_{pn} \\ \underline{Z}_{pn} & \underline{Z}_{pn} & \underline{Z}_{pn} & \underline{Z}_n \end{bmatrix} \qquad [\underline{\boldsymbol{Y}}_{abcn}] = \begin{bmatrix} \underline{Y}_s & \underline{Y}_m & \underline{Y}_m & \underline{Y}_{pn} \\ \underline{Y}_m & \underline{Y}_s & \underline{Y}_m & \underline{Y}_{pn} \\ \underline{Y}_m & \underline{Y}_m & \underline{Y}_s & \underline{Y}_{pn} \\ \underline{Y}_{pn} & \underline{Y}_{pn} & \underline{Y}_{pn} & \underline{Y}_n \end{bmatrix} \tag{22}$$

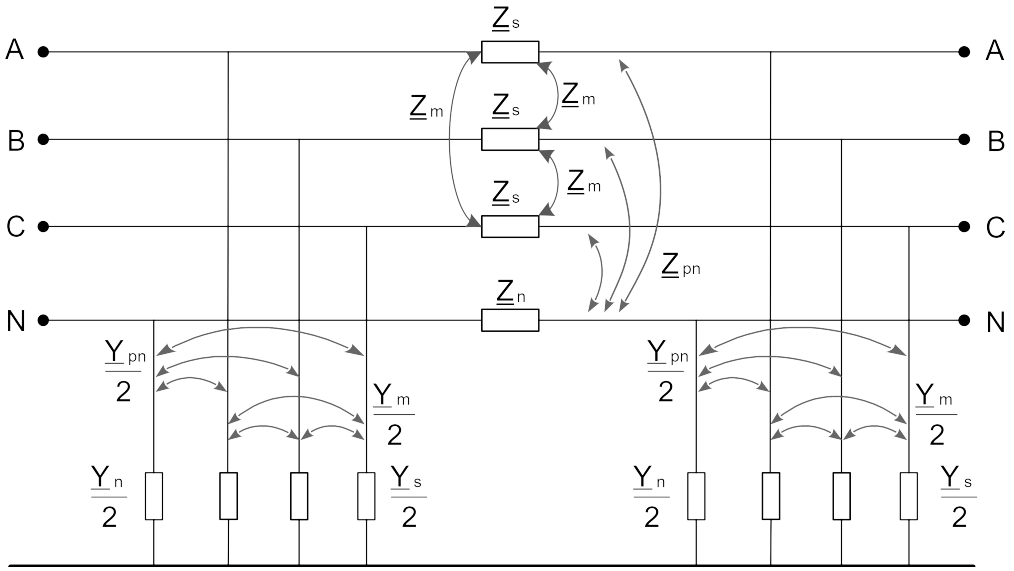


Figure 2.7: Equivalent circuit for the 3-phase line with neutral conductor

The input parameters of the model are the positive and zero sequence impedances $\underline{Z}_1$, $\underline{Z}_0$. The positive and zero sequence admittance, $\underline{Y}_1$ and $\underline{Y}_0$, the self- and mutual impedance $\underline{Z}_n$, $\underline{Z}_{pn}$ and the admittance for the neutral conductor $\underline{Y}_n$, $\underline{Y}_{pn}$ are as listed in Table A.1 (Input parameter of the line type (*TypLne*)) of annex A.

The values $\underline{Z}_n$, $\underline{Z}_{pn}$, $\underline{Y}_p$ and $\underline{Y}_{pn}$ of the neutral conductor can be directly used in (22). The self- and mutual impedance are determined like for the three phase line model (see equations (20) and (21)).

2.1.7 4-Wire Model Data Conversion

The self- and mutual impedances and admittances are not always available in the format required by the line type (*TypLne*). The following subsections guide the reader on how to convert commonly available measurement data into the required input parameters.

2.1.7.1 Measurement between phase A and phase B wire

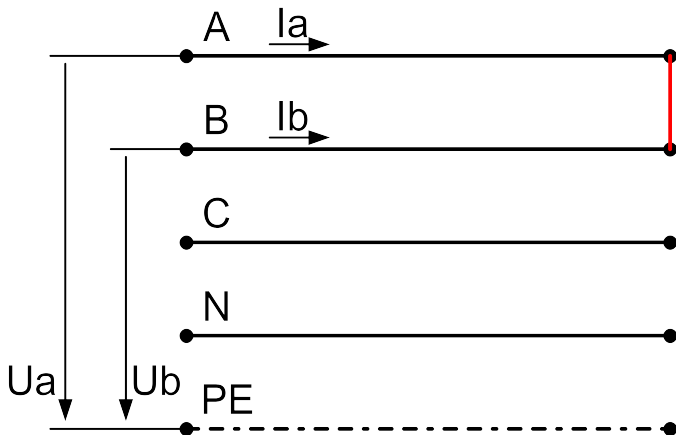


Figure 2.8: Phase to phase measurement loop

As per the impedance matrix in (22):

$$\begin{bmatrix} \underline{U}_a \\ \underline{U}_b \\ \underline{U}_c \\ \underline{U}_n \end{bmatrix} = \begin{bmatrix} \underline{Z}_s & \underline{Z}_m & \underline{Z}_m & \underline{Z}_{pn} \\ \underline{Z}_m & \underline{Z}_s & \underline{Z}_m & \underline{Z}_{pn} \\ \underline{Z}_m & \underline{Z}_m & \underline{Z}_s & \underline{Z}_{pn} \\ \underline{Z}_{pn} & \underline{Z}_{pn} & \underline{Z}_{pn} & \underline{Z}_n \end{bmatrix} \begin{bmatrix} \underline{I}_a \\ \underline{I}_b \\ \underline{I}_c \\ \underline{I}_n \end{bmatrix} \tag{23}$$

thus:

$$\underline{U}_a = \underline{Z}_s \cdot \underline{I}_a + \underline{Z}_m \cdot \underline{I}_b$$
$$\underline{U}_b = \underline{Z}_m \cdot \underline{I}_a + \underline{Z}_s \cdot \underline{I}_b$$

According to the measurement in Figure 2.8:

$$\underline{I}_a = (-\underline{I}_b)$$
$$U_a - U_b = 2I_a \cdot (\underline{Z}_s - \underline{Z}_m)$$

the positive sequence impedance can be derived as follows:

$$\underline{Z}_1 = R_1 + jX_1 = \frac{1}{2}\frac{\underline{U}_a - \underline{U}_b}{\underline{I}_a} \tag{24}$$

2.1.7.2 Measurement between neutral and PE (earth) wire

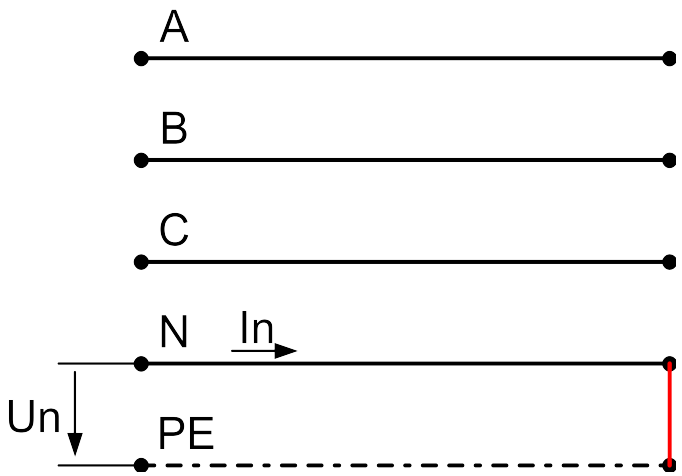


Figure 2.9: Neutral to ground measurement loop

Again from (23) now with Ia=Ib=Ic=0:

$$U_n = \underline{Z}_n \cdot \underline{I}_n$$

$$\underline{Z}_n = \frac{\underline{U}_n}{\underline{I}_n} \tag{25}$$

Note that $\underline{Z}_n$ represents the impedance of the neutral conductor, $\underline{Z}_{Neutral}$, plus the impedance of the earth-return path, $\underline{Z}_{earth}$.

2.1.7.3 Measurement between phase and PE (earth) wire

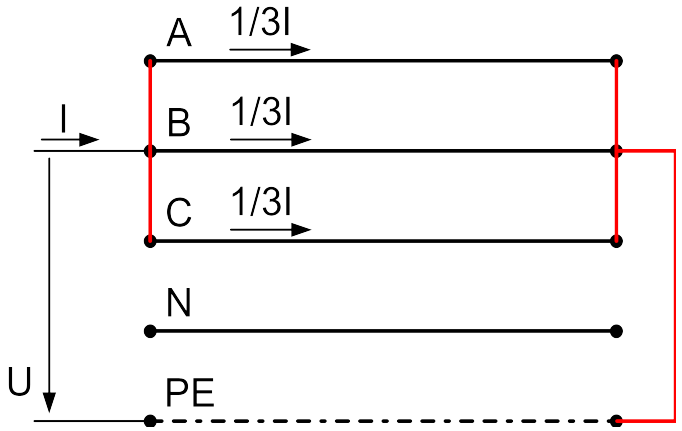


Figure 2.10: Phase – PE (earth wire) measurement

From equation (23) with In = 0

$$\underline{U} = \frac{1}{3}\left(\underline{Z}_s \cdot \underline{I} + \underline{Z}_m \cdot \underline{I} + \underline{Z}_m \cdot \underline{I}\right)$$

$$\underline{U} = \left(\frac{1}{3}\underline{I}\right) \cdot \underline{Z}_0$$

the zero-sequence impedance is:

$$\underline{Z}_0 = \frac{3 \cdot \underline{U}}{\underline{I}} \tag{26}$$

2.1.7.4 Measurement between phase and neutral wire

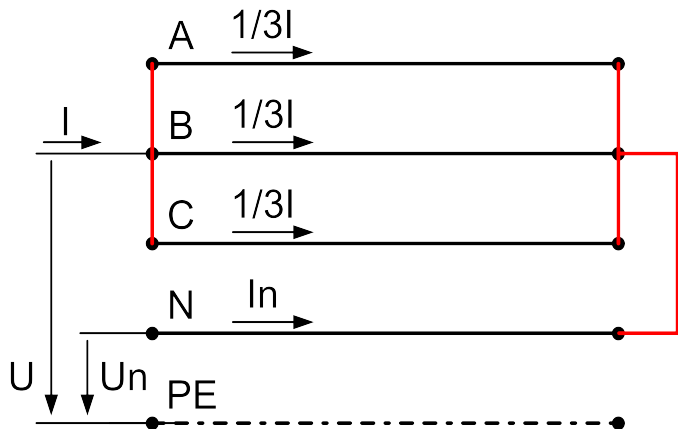


Figure 2.11: Phase – neutral measurement

From equation (23):

$$\underline{U} = \frac{1}{3}\left(\underline{Z}_s \cdot \underline{I} + \underline{Z}_m \cdot \underline{I} + \underline{Z}_m \cdot \underline{I}\right) + \underline{Z}_{pn} \cdot \underline{I}_n$$

$$\underline{U}_n = \frac{1}{3} \cdot \underline{I} \cdot \left(\underline{Z}_{pn} + \underline{Z}_{pn} + \underline{Z}_{pn}\right) + \underline{Z}_n \cdot \underline{I}_n$$

With $\underline{I}_n = -\underline{I}$:

$$\underline{U} = \frac{1}{3}\left(\underline{Z}_s \cdot \underline{I} + \underline{Z}_m \cdot \underline{I} + \underline{Z}_m \cdot \underline{I}\right) - \underline{Z}_{pn} \cdot \underline{I}$$

$$\underline{U}_n = \frac{1}{3} \cdot \underline{I} \cdot \left(\underline{Z}_{pn} + \underline{Z}_{pn} + \underline{Z}_{pn}\right) - \underline{Z}_n \cdot \underline{I}$$

And according to equations (20) and (21):

$$\underline{U} = \frac{1}{3}\underline{Z}_0 \cdot \underline{I} - \underline{Z}_{pn} \cdot \underline{I}$$

$$\underline{U}_n = \underline{I} \cdot \left(\underline{Z}_{pn} - \underline{Z}_n\right)$$

Subtracting these equations:

$$\underline{Z}_{0,PH-N} = \frac{3\left(\underline{U} - \underline{U}_n\right)}{\underline{I}} = \underline{Z}_0 - 6 \cdot \underline{Z}_{pn} + 3 \cdot \underline{Z}_n$$

$\underline{Z}_{0,PH-N}$ is commonly referred to as the zero-sequence impedance between with return over the neutral conductor. With $\underline{Z}_n$ given by (25) the mutual impedance between phase and neutral conductors is:

$$\underline{Z}_{pn} = \frac{\underline{Z}_0 + 3 \cdot \underline{Z}_n - \underline{Z}_{0,PH-N}}{6} \tag{27}$$

2.1.7.5 Data conversion without N-PE measurement

If the measurement between the neutral and the PE (earth) wire does not exist the following simplification can be made:

Phase-neutral loop:

$$\underline{Z}_{0,PH-N} = \underline{Z}_1 + 3 \cdot \underline{Z}_{Neutral} \tag{28}$$

Phase-ground loop:

$$\underline{Z}_0 = \underline{Z}_1 + 3 \cdot \underline{Z}_{Earth} \tag{29}$$

Thus the neutral-earth loop impedance:

$$\underline{Z}_{N-E} = \underline{Z}_{Neutral} + \underline{Z}_{Earth}$$

$$\underline{Z}_{N-E} = \underline{Z}_n = \frac{\underline{Z}_{0,PH-N} + \underline{Z}_0 - 2 \cdot \underline{Z}_1}{3} \tag{30}$$

And the phase-neutral mutual impedance:

$$\underline{Z}_{pn} = \frac{\underline{Z}_0 - \underline{Z}_1}{3} \tag{31}$$

2.2 Distributed Parameter Model

In addition to the lumped parameter models described in previous sections, *PowerFactory* also supports distributed parameters models for 3-phase line circuits. This kind of model accounts for the distributed nature of the line parameters and should therefore be the preferred option for the modelling of long lines. For short lines, the lumped parameter models discussed in the previous sections should provide sufficiently accurate solutions.

A line is considered to be long when its physical length is of the same order of magnitude as the length of wave of the voltage/current at the frequency under consideration (e.g. system frequency for the Load Flow calculation). Note that for increasing frequencies, and hence for example for harmonic Load Flow calculations, the higher the frequency the lower the length of wave, so that even a physically short line may need to be treated as a long line, and therefore be represented using distributed parameters.

To select a distributed parameter model, the corresponding model option on the Basic Data page of the line element (*ElmLne*) or line coupling (*ElmTow*) should be selected.

2.2.1 General Formulation

Equations (32) and (33) describe the incremental transmission line model in the frequency domain of an elemental length Δx depicted in Figure 2.12.

$$\frac{\partial}{\partial x} V = I(x) \cdot Z' \tag{32}$$

$$\frac{\partial}{\partial x} I = V(x) \cdot Y' \tag{33}$$

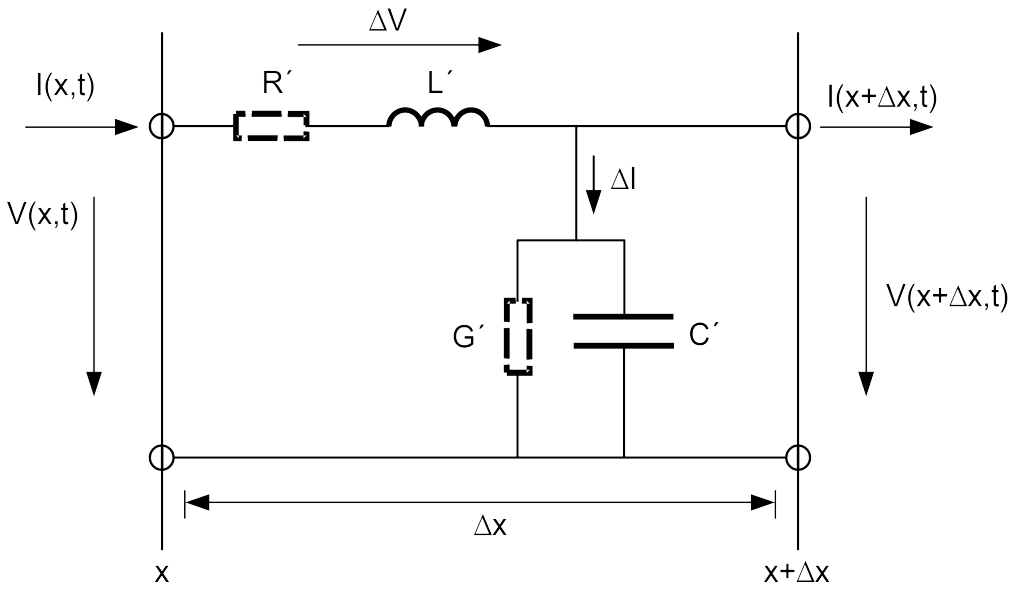


Figure 2.12: Incremental model for a line of elemental length

After taking the second derivatives of (32) and (33) with respect to x and rearranging the equations to separate the voltage from the current magnitudes, the system of differential equations can be rewritten as:

$$\begin{aligned} \tfrac{\partial^2}{\partial x^2} V &= Z' \cdot Y' \cdot V(x) \\ \tfrac{\partial^2}{\partial x^2} I &= Z' \cdot Y' \cdot I(x) \end{aligned} \tag{34}$$

The general solution is of the form:

$$\begin{aligned} U(x) &= K_1 \cdot e^{\gamma \cdot x} + K_2 \cdot e^{-\gamma \cdot x} \\ Z_C \cdot I(x) &= -K_1 \cdot e^{\gamma \cdot x} + K_2 \cdot e^{-\gamma \cdot x} \end{aligned} \tag{35}$$

with

$$Z_C = \sqrt{\frac{Z'}{Y'}} \tag{36}$$

$$\gamma = \sqrt{Z' \cdot Y'} = \alpha + j\beta \tag{37}$$

Both the surge (or characteristic) impedance Z_C and the propagation factor γ are frequency-dependent and uniquely characterise the behaviour of the transmission line. Further details regarding the derivation of these equations can be found in [1], [3].

The integration constants K_1 and K_2 in (35) are determined from the border conditions at either the receiving or the sending end of the line. According the sign convention in Figure 2.13, the particular solution of (35) results:

$$\begin{bmatrix} U_r \\ I_r \end{bmatrix} = \begin{bmatrix} \cosh\gamma \cdot l & -Z_C \cdot \sinh\gamma \cdot l \\ \frac{1}{Z_C} \cdot \sinh\gamma \cdot l & -\cosh\gamma \cdot l \end{bmatrix} \cdot \begin{bmatrix} U_s \\ I_s \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \cdot \begin{bmatrix} U_s \\ I_s \end{bmatrix} \tag{38}$$

and therefore the impedance and admittance of the equivalent circuit are:

$$\begin{aligned} Z &= Z_C \cdot \sinh\gamma \cdot l = Z' \cdot l \cdot \tfrac{\sinh\gamma \cdot l}{\gamma \cdot l} \\ Y &= \tfrac{\cosh\gamma \cdot l - 1}{Z_C \cdot \sinh\gamma \cdot l} = \tfrac{1}{2} \cdot Y' \cdot l \cdot \tfrac{\tanh\left(\frac{\gamma \cdot l}{2}\right)}{\frac{\gamma \cdot l}{2}} \end{aligned} \tag{39}$$

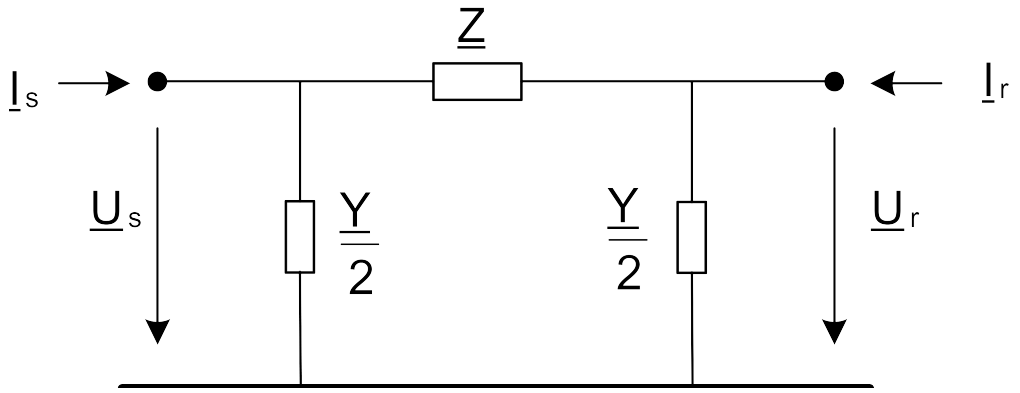


Figure 2.13: Equivalent pi-circuit for the line with distributed parameters in the frequency domain

It should be noted that Z and Y in Figure 2.13 are frequency-dependent parameters as both the surge impedance, Z_C, and the propagation factor, γ, are functions of the frequency.

Series expansion

The lumped parameter model described by (11) in Section 2.1.1 is a simplified model of the distributed parameters model. Taking a series expansion of the hyperbolic functions in (39) gives:

$$\cosh\vartheta = 1 + \frac{1}{2} \cdot \vartheta^2 + \frac{1}{24} \cdot \vartheta^4 + \frac{1}{720} \cdot \vartheta^6 + \ldots$$

$$\frac{\sinh\vartheta}{\vartheta} = 1 + \frac{1}{6} \cdot \vartheta^2 + \frac{1}{120} \cdot \vartheta^4 + \frac{1}{5040} \cdot \vartheta^6 + \ldots$$

Using $\vartheta = \gamma \cdot l = \sqrt{Z' \cdot Y'} \cdot l$ *A* and *B* in (38) can be expanded as follows:

$$A = \cosh\gamma \cdot l = 1 + \frac{1}{2} \cdot Z' \cdot Y' \cdot l^2 + \frac{1}{24} \cdot (Z' \cdot Y')^2 \cdot l^4 + \ldots$$

$$B = Z' \cdot l \cdot \left(\frac{\sinh\gamma \cdot l}{\gamma \cdot l}\right) = Z' \cdot l \cdot \left[1 + \frac{1}{6} \cdot Z' \cdot Y' \cdot l^2 + \frac{1}{120} \cdot (Z' \cdot Y')^2 \cdot l^4 + \ldots\right]$$

Considering up to the second order terms, equations (39) of the distributed parameter model go into equations (9) of the lumped parameter model:

$$Z = B = Z' \cdot l = R' \cdot l + j\omega \cdot L' \cdot l$$

$$Y = \frac{A-1}{B} = \frac{1 + \frac{1}{2} \cdot Z' \cdot Y' \cdot l^2}{Z' \cdot l} = \frac{1}{2} \cdot Y' \cdot l = \frac{1}{2} \cdot (G' \cdot l + j\omega \cdot C' \cdot l)$$

The accuracy of the lumped model therefore depends on the weight of truncated terms in the series expansion, which in turns depends on the factor $f \cdot l$ (frequency multiplied by length). For overhead lines with a length less than 250 km and at system frequency, this approximation is sufficient and the error is negligible. For longer lines or higher frequencies, a distributed parameter model will provide a more accurate solution.

Longer lines can be alternatively modelled by cascading line sections. In general, the longer the line or the higher the frequency, the more line sections are required in order to obtain the same accuracy. Increasing the number of line sections to infinity will turn the lumped parameter model into the distributed parameters model discussed previously.

3 Load Flow Analysis

3.1 Temperature Dependency of Line Resistance

For a line type a temperature dependency can be defined either via the conductor material:

- Aluminium ($\alpha = 0.00403$ 1/K)
- Copper ($\alpha = 0.00393$ 1/K)
- Aldrey ($\alpha = 0.00360$ 1/K)
- Aluminium-Steel ($\alpha = 0.00403$ 1/K)
- Aldrey-Steel ($\alpha = 0.00360$ 1/K)

with a temperature coefficient α or with a resistance at max. operating temperature ($tmax$ in degC and $rline_{tmax}$ in Ohm/km)

Depending on the temperature setting e.g. in the load flow command, frequency sweep the positive sequence resistance is scaled as follow when using a line type (*TypLne*):

$$R_1^{'} = rline \cdot (1 + \alpha \cdot (temp - 20) \tag{40}$$

Note: The zero sequence resistance is not adapted.

At which temperature the resistance is modified depends on the setting e.g. of the load flow command or/and e.g. the entered operating temperature at the line.

- at 20 degC
- at max. operating temperature ($tmax$ in line type, conductor type, cable type)
- at operating temperature (Top entered in the line model)
- at temperature entered in load flow or frequency sweep command

Note: When using a tower model, and the temperature is configured as ”at operating temperature”, the earth conductor is still applied by the entered resistance at 20 degC. For lines using a cable type (*TypCabsys*), or a cable system (*TypCabsys*), the conductor, sheath and armour the corresponding temperature is applied.

3.2 AC Model Calculation Parameters

Loading

The loading of the line/cable is calculated as follows:

$$loading = max\left(\frac{|I_{busi}|}{I_{nom(busi)}}, \frac{|I_{busj}|}{I_{nom(busj)}}\right) \cdot 100 \quad \% \tag{41}$$

where:

- $I_{nom(busi)}$ is nominal current of the line/cable for terminal i in kA

- $I_{nom(busj)}$ is nominal current of the line/cable for terminal j in kA
- I_{busi} is magnitude of the current at terminal i
- I_{busj} is magnitude of the current at terminal j

If no thermal ratings object is defined, the nominal current is:

$I'_{nom(busi)} = I'_{nom(busj)} = I_r \cdot fline$ (see equation 1 or 2).

where $fline$ is the derating factor of the line (if a thermal ratings object is selected, $fline$ is not used in the calculation).

If a thermal ratings object is used, the nominal currents are calculated using the parameter $ContRating$ and $U_{n(busi)}$, $U_{n(busj)}$ (nominal voltage in kV of the connected terminal i and j) as:

- if the continuous rating is entered in MVA:
  - 3-phase AC:

$$I'_{nom(busi)} = ContRating / \left(\sqrt{3} \cdot U_{n(busi)}\right) \cdot nlnum \quad \text{kA}$$
$$I'_{nom(busj)} = ContRating / \left(\sqrt{3} \cdot U_{n(busj)}\right) \cdot nlnum \quad \text{kA}$$

  - 2-phase AC:

$$I'_{nom(busi)} = ContRating / U_{n(busi)} \cdot nlnum \quad \text{kA}$$
$$I'_{nom(busj)} = ContRating / U_{n(busj)} \cdot nlnum \quad \text{kA}$$

  - 1-phase AC:

$$I'_{nom(busi)} = ContRating \cdot \sqrt{3} / U_{n(busi)} \cdot nlnum \quad \text{kA}$$
$$I'_{nom(busj)} = ContRating \cdot \sqrt{3} / U_{n(busj)} \cdot nlnum \quad \text{kA}$$

  - 1-phase connected on a AC/BI system:

$$I'_{nom(busi)} = ContRating \cdot 2 / U_{n(busi)} \cdot nlnum \quad \text{kA}$$
$$I'_{nom(busj)} = ContRating \cdot 2 / U_{n(busj)} \cdot nlnum \quad \text{kA}$$

- if the continuous rating is entered in kA:

$$I'_{nom(busi)} = I'_{nom(busi)} = ContRating \cdot nlnum \quad \text{kA}$$

- if the continuous rating is entered in %:

$$I'_{nom(busi)} = I'_{nom(busi)} = ContRating / 100 \cdot I_r \quad \text{kA}$$

The nominal currents are generally the same ($I'_{nom(busi)} = I'_{nom(busj)}$), but not if the line/cable is connected between different voltage levels ($U_{n(busi)} \neq U_{n(busi)}$).

Further details regarding thermal ratings object (*IntThrating*), see Technical Reference of the Thermal Ratings Definition

Note: In case if the nominal current is greater than the max. permissible current, the max. permissible current ($Imaxlim$ in kA, see Load Flow page of the line) is used for the nominal current.

- $I_{nom(busi)} = min(I'_{nom(busi)},\ Imaxlim \cdot nlnum)$
- $I_{nom(busj)} = min(I'_{nom(busj)},\ Imaxlim \cdot nlnum)$

For an unbalanced load flow calculation the highest current of all phases is used.

Losses

The losses are calculated as:

$$Losses = (P_{busi} + P_{busj}) \cdot 1000 \quad \text{kW} \tag{42}$$

where P_{busi} and P_{busj} are calculation parameters available for the branch element.

The total losses are obtained as:

$$\begin{aligned} Ploss &= P_{busi} + P_{busj} - P_{lineloads} \quad \text{MW} \\ Qloss &= Q_{busi} + Q_{busj} - Q_{lineloads} \quad \text{Mvar} \end{aligned} \tag{43}$$

where $P_{lineloads}$ and $Q_{lineloads}$ are the active and reactive power of the line loads.

The load losses are obtained using the no-load loss parameters:

$$\begin{aligned} Plossld &= Ploss - Plossnld \quad \text{MW} \\ Qlossld &= Qloss - Qlossnld \quad \text{Mvar} \end{aligned} \tag{44}$$

The no-load losses are calculated using the *Active no-load losses* $Gload$ and the *Capacitative loading* $Cload$ as follows:

$$\begin{aligned} \underline{S}_{lossnld} &= \underline{u}_{busi} \cdot (\underline{y}_{line} \cdot \underline{u}_{busi})^* + \underline{u}'_{busi} \cdot (\underline{y}_{line} \cdot \underline{u}'_{busj})^* \quad \text{MVA} \\ Gload &= \Re(\underline{S}_{lossnld}) \cdot 1000 \quad \text{kW} \\ Cload &= -\Im(\underline{S}_{lossnld}) \quad \text{Mvar} \\ Plossnld &= Gload/1000 \quad \text{MW} \\ Qlossnld &= -Cload \quad \text{Mvar} \end{aligned} \tag{45}$$

where $\underline{u}_{busi}$ and $\underline{u}_{busj}$ are the terminal voltages where the line is connected in p.u., $\underline{u}'_{busj} = \underline{u}_{busj} \cdot U_{n(busj)}/U_{n(busi)}$ (nominal voltages of the connected busbars) and $\underline{y}_{line}$ is the complex admittance of the line.

Voltage Drop

The voltage drop and voltage angle drops are calculated as:

$$\begin{aligned} du &= |u_{busi}| - |u_{busj}| \quad \text{p.u.} \\ dupc &= du \cdot 100\ \% \\ dphiu &= \phi_{u_busi} - \phi_{u_busj} \quad \text{deg} \end{aligned} \tag{46}$$

and the positive-sequence voltage drop and positive-sequence voltage angle drop are calculated as:

$$\begin{aligned} d1u &= |u1_{busi}| - |u1_{busj}| \quad \text{p.u.} \\ du1pc &= du1 \cdot 100\ \% \\ dphiu1 &= \phi_{u1,busi} - \phi_{u1,busj} \quad \text{deg} \end{aligned} \tag{47}$$

where u_{busi} and u_{busj} are the amplitudes of the corresponding terminal voltage in p.u. based on the rated voltage of the terminal, ϕ_{u_busi} and ϕ_{u_busj} are the terminal voltage angles in deg. For an unbalanced load flow du, $dupc$, and $dphiu$ are available per phase (e.g. $c : dupc : B$).

3.3 AC Model Calculation Parameters for Linear DC Load Flow

Loading

The loading of the line/cable is calculated as follows:

$$loading = max\left(\frac{|P_{busi}|}{P_{nom(busi)}}, \frac{|P_{busj}|}{P_{nom(busj)}}\right) \cdot 100 \quad \%$$

where:

- P_{busi} : Active power at terminal i
- P_{busj} : Active power at terminal j
- $P_{nom(busi)}$: Nominal power at terminal i
- $P_{nom(busj)}$: Nominal power at terminal j

The nominal power is determined as follow when no thermal ratings object is defined:

- $P'_{nom(busi)} = \sqrt{3} \cdot U_{n(busi)} \cdot I_r \cdot fline$
- $P'_{nom(busj)} = \sqrt{3} \cdot U_{n(busj)} \cdot I_r \cdot fline$

with the rated current of the line/cable I_r (see equation 1 or 2).

where $fline$ is the derating factor of the line (if a thermal ratings object is selected, $fline$ is not used in the calculation).

$U_{n(busi)}$, $U_{n(busj)}$ is the nominal voltage in kV of the connected terminals.

If a thermal ratings object is used, the nominal power is calculated using the parameter $ContRating$ and $U_{n(busi)}$ and $U_{n(busj)}$ (nominal voltage in kV of the connected terminal i and j) as:

- if the continuous rating is entered in MVA:

$$P'_{nom(busi)} = P'_{nom(busj)} = ContRating \cdot nlnum \quad \text{MW}$$

- if the continuous rating is entered in kA:

$$P'_{nom(busi)} = \sqrt{3} \cdot U_{n(busi)} \cdot ContRating \cdot nlnum \quad \text{MW}$$
$$P'_{nom(busj)} = \sqrt{3} \cdot U_{n(busj)} \cdot ContRating \cdot nlnum \quad \text{MW}$$

- if the continuous rating is entered in %:

$$P'_{nom(busi)} = \sqrt{3} \cdot U_{n(busi)} \cdot ContRating/100 \cdot I_r \quad \text{MW}$$
$$P'_{nom(busj)} = \sqrt{3} \cdot U_{n(busj)} \cdot ContRating/100 \cdot I_r \quad \text{MW}$$

The nominal powers are generally the same ($P'_{nom(busi)} = P'_{nom(busj)}$), but not if the line/cable is connected between different voltage levels ($U_{n(busi)} \neq U_{n(busi)}$).

Further details regarding thermal ratings object (*IntThrating*), see Technical Reference of the Thermal Ratings Definition

Note: In case if the nominal power is greater than the max. permissible power (based on the parameter $Imaxlim$ in kA, see Load Flow page of the line), the max. permissible power is used for the nominal power:

- $P_{nom(busi)} = min(P'_{nom(busi)},\ \sqrt{3} \cdot U_{n(busi)} \cdot Imaxlim \cdot nlnum)$
- $P_{nom(busj)} = min(P'_{nom(busj)},\ \sqrt{3} \cdot U_{n(busj)} \cdot Imaxlim \cdot nlnum)$

Losses

Losses are not calculated in the linear DC Load Flow.

3.4 DC Model Calculation Parameters

Loading

The loading of the line/cable is calculated as follows:

$$loading = max\left(\frac{|I_{busi}|}{I_{nom(busi)}},\ \frac{|I_{busj}|}{I_{nom(busj)}}\right) \cdot 100 \quad \%$$

- $I_{nom(busi)}$ is nominal current of the line/cable for terminal i in kA
- $I_{nom(busj)}$ is nominal current of the line/cable for terminal j in kA
- I_{busi} DC current at terminal i
- I_{busj} DC current at terminal j

If no thermal ratings object is defined, the nominal current is:

$I'_{nom(busi)} = I'_{nom(busj)} = I_r \cdot fline$ (see equation 1 or 2).

where $fline$ is the derating factor of the line (if a thermal ratings object is selected, $fline$ is not used in the calculation).

If a thermal ratings object is used, the nominal currents are calculated using the parameter $ContRating$ and $U_{n(busi)}$,$U_{n(busj)}$ (nominal voltage in kV of the connected terminal i and j) as:

- if the continuous rating is entered in MVA:

$$I'_{nom(busi)} = ContRating/U_{n(busi)} \cdot nlnum \quad \text{kA}$$
$$I'_{nom(busj)} = ContRating/U_{n(busj)} \cdot nlnum \quad \text{kA}$$

- if the continuous rating is entered in kA:

$$I'_{nom(busi)} = I'_{nom(busj)} = ContRating \cdot nlnum \quad \text{kA}$$

- if the continuous rating is entered in %:

$$I'_{nom(busi)} = I'_{nom(busj)} = ContRating/100 \cdot I_r \quad \text{kA}$$

The nominal currents are generally the same ($I'_{nom(busi)} = I'_{nom(busj)}$), but not if the line/cable is connected between different voltage levels ($U_{n(busi)} \neq U_{n(busi)}$).

Further details regarding thermal ratings object (*IntThrating*), see Technical Reference of the Thermal Ratings Definition

Note: In case if the nominal current is greater than the max. permissible current, the max. permissible current ($Imaxlim$ in kA, see Load Flow page of the line) is used for the nominal current.

- $I_{nom(busi)} = min(I'_{nom(busi)},\ Imaxlim \cdot nlnum)$
- $I_{nom(busj)} = min(I'_{nom(busj)},\ Imaxlim \cdot nlnum)$

Losses

The losses are calculated as:

$$Losses = (P_{busi} + P_{busj}) \cdot 1000 \quad \text{kW} \tag{48}$$

where P_{busi} and P_{busj} are calculation parameters available for the branch element.

The total losses are obtained as:

$$\begin{aligned} Ploss &= P_{busi} + P_{busj} \quad \text{MW} \\ Qloss &= 0 \quad \text{Mvar} \end{aligned} \tag{49}$$

The load losses are obtained using the no-load loss parameters:

$$\begin{aligned} Plossld &= Ploss - Plossnld \quad \text{MW} \\ Qlossld &= 0 \quad \text{Mvar} \end{aligned} \tag{50}$$

The no-load losses are calculated using the *Active no-load losses* $Gload$ as follows:

$$\begin{aligned} Gload &= (1/2 \cdot glne \cdot U_{busi}^2 + 1/2 \cdot glne \cdot U_{busj}^2) \cdot 1000 \quad \text{kW} \\ Cload &= 0 \quad \text{Mvar} \\ Plossnld &= Gload/1000 \quad \text{MW} \\ Qlossnld &= 0 \quad \text{Mvar} \end{aligned} \tag{51}$$

where $glne$ is the admittance (calculation parameter) of the line and U_{busi} and U_{busj} are the terminal DC voltages where the line is connected.

3.5 Consideration of Line Shunt Compensation

For load flow calculations, line shunt compensation is modelled as a shunt admittance at each end, *i,j*, of the line: $\underline{y}_{i_{comp}}$ and $\underline{y}_{j_{comp}}$.

These shunt admittances are used to calculate the shunt compensation currents, i_{comp}, as follows:

$$\begin{aligned} \underline{i}_{comp_i} &= \underline{i}_i - \underline{y}_{i_{comp}} \cdot \underline{u}_i \\ \underline{i}_{comp_j} &= \underline{i}_j - \underline{y}_{j_{comp}} \cdot \underline{u}_j \end{aligned} \tag{52}$$

For unbalanced load flow calculations, the voltages are first transformed to the sequence system and then multiplied by the positive and zero sequence admittances accordingly. Transformation back to the phase domain is then used to obtain the phase shunt compensation currents.

4 Short-Circuit

For short-circuit calculations, transmission lines can be represented using either the lumped parameter model or the distributed parameter model. These are described in Sections 2.1 and 2.2, respectively.

Note: The distributed parameter model is not used by the VDE and ANSI methods. The *Complete* method will only use the distributed parameter model if the Short-Circuit command options are configured as follows:

- *Load flow initialisation* is selected; or
- *Ignore positive sequence data* → *Capacitance of lines* is deselected

In addition, the distributed parameter line will not be used during the R/X calculation.

4.1 Zero sequence mutual impedance

With the zero sequence mutual impedance objects two or more lines can be coupled to defined the zero sequence coupling impedances between lines and is only relevant for unbalanced calculation method.

Note: The zero sequence mutual impedance is only supported for three phase lines or line sections (without neutral wire).

The zero sequence mutual impedance object an be defined for lines (*ElmLne*) or line sections (*ElmLnesec*). In case if a line is select also the from, to length can be defined. An example is shown in the following Figure 4.1.

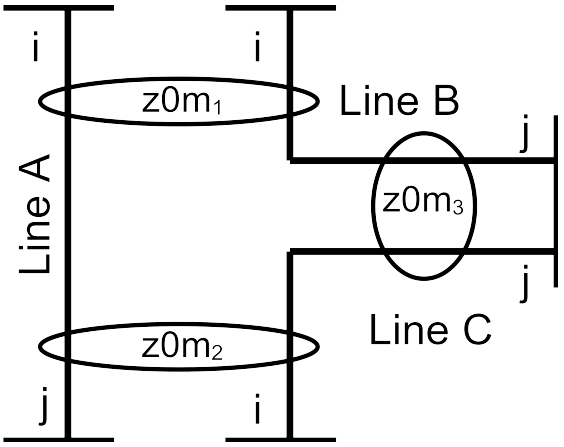


Figure 4.1: Example for zero sequence mutual impedance coupling

For the shown example are three zero sequence mutual impedance objects required $z0m_1$,$z0m_2$ and $z0m_3$.

- $z0m_1$ is configured as follow:
  - Line 1: *Line A*; Start position = 0 %; End position = 40 %

  - Line 2: *Line A*; Start position = 0 %; End position = 30 %
  - Orientation = same direction

- $z0m_2$ is configured as follow:
  - Line 1: *Line A*; Start position = 60 %; End position = 100 %
  - Line 2: *Line C*; Start position = 0 %; End position = 40 %
  - Orientation = opposite direction

- $z0m_3$ is configured as follow:
  - Line 1: *Line B*; Start position = 30 %; End position = 100 %
  - Line 2: *Line C*; Start position = 30 %; End position = 100 %
  - Orientation = same direction

The zero sequence impedance is entered in Ω, the from and to position in percent of the line length. Further the orientation can be defined e.g. in case if two lines are coupled in opposite direction (see Figure 4.2).

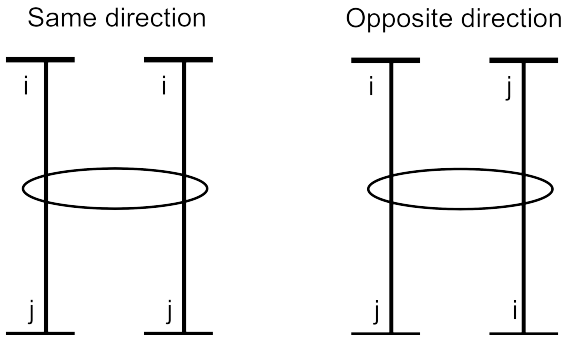


Figure 4.2: Orientation of zero sequence mutual impedance

All zero sequence mutual impedance object will be collected and a zero sequence admittance matrix will be created

$$|Z0| = \begin{bmatrix} Z0_{11} & Z0m_{12} & Z0m_{13} & ... & Z0m_{1n} \\ Z0m_{21} & Z0_{22} & Z0m_{23} & ... & Z0m_{2n} \\ Z0m_{31} & Z0m_{32} & Z0_{33} & ... & Z0m_{3n} \\ ... & ... & ... & ... & ... \\ Z0m_{n1} & Z0m_{n2} & Z0_{n3} & ... & Z0_{nn} \end{bmatrix} \quad (53)$$

where

- $Z0_{11}$,$Z0_{22}$, $Z0_{nn}$: zero sequence impedance of the corresponding line 1,2, ... n
- $Z0m_{12}$, $Z0m_{13}$, $Z0m_{32}$ ...: zero sequence mutual impedance of the corresponding coupling objects

The zero sequence impedance matrix will be inverted to get a zero sequence admittance matrix:

$$|Y0| = |Z0|^{-1} \quad (54)$$

The zero sequence current for a line is than calculated as follow:

$$I0 = |Y0| \cdot |dU0| \tag{55}$$

where $dU0$ is the zero sequence voltage difference of the coupled line. e.g. for line 1

$$dU0_1 = U0_{i,1} - U0_{j,1} \qquad \text{in case of same direction orientation}$$
$$dU0_1 = U0_{j,1} - U0_{i,1} \qquad \text{in case of opposite direction orientation}$$

In case of a short-circuit at the line, the corresponding zero sequence mutual impedance objects are depending on the short-circuit location internally split. The zero sequence current of side i and j are than calculated accordingly.

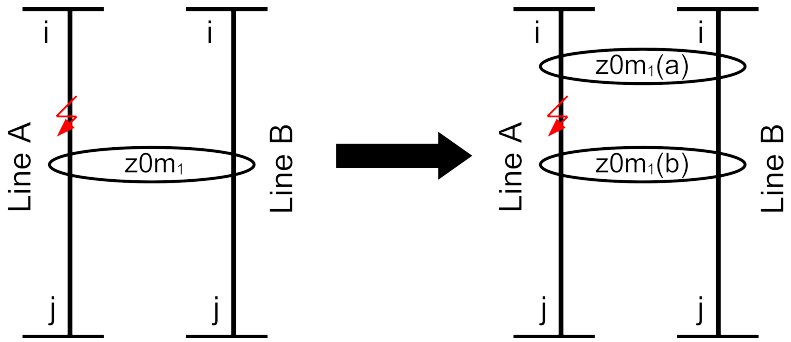


Figure 4.3: Short-circuit at line with zero sequence mutual impedance

The zero sequence mutual impedance is also split according to the short-circuit location into two mutual coupling objects.

Note: Only the short-circuit line is split into two PI-Models.

4.2 Consideration of Line Shunt Compensation

Line shunt compensation is ignored by the balanced IEC and ANSI methods, and is also ignored for the R/X ratio calculation (also complete method). When using the *Complete* method, the positive line shunt admittance is considered as in the load flow or RMS simulation (see Section 3.5).

For unbalanced short-circuit calculations, only the zero sequence admittance is considered by the IEC and ANSI methods. When the R/X ratio is calculated (for the *Complete* method as well) the imaginary part of the zero sequence admittance is ignored.

5 Harmonics Analysis

For harmonics analysis, transmission lines can be represented using either the lumped parameter model or the distributed parameter model. These are described in Sections 2.1 and 2.2, respectively.

5.1 Zero sequence mutual impedance

In case of a zero sequence mutual impedance is defined (see chapter 4.1).

The zero sequence mutual reactance is scaled by the harmonic order.

5.2 Consideration of Line Shunt Compensation

The line compensation is similar as in the EMT simulation considered depending on if the compensation is a reactor or capacitor.

5.2.0.1 Shunt Reactor

For line compensation that is modelled as a shunt reactor, the frequency dependency is applied as follows:

$$\begin{aligned} \underline{Z}_{1_{comp}} &= \frac{1}{\underline{Y}_{1_{comp}}} \\ \underline{Z}_{0_{comp}} &= \frac{1}{\underline{Y}_{0_{comp}}} \end{aligned} \tag{56}$$

$$\begin{aligned} \Im(\underline{Z}_{1_{comp}}) &= \Im(\underline{Z}_{1_{comp}}) \cdot f_h \\ \Im(\underline{Z}_{0_{comp}}) &= \Im(\underline{Z}_{0_{comp}}) \cdot f_h \end{aligned} \tag{57}$$

where f_h is the harmonic order.

5.2.0.2 Shunt Capacitor

For line compensation that is modelled as a shunt capacitor, the frequency dependency is applied as follows:

$$\begin{aligned} \Im(\underline{Y}_{1_{comp}}) &= \Im(\underline{Y}_{1_{comp}}) \cdot f_h \\ \Im(\underline{Y}_{0_{comp}}) &= \Im(\underline{Y}_{0_{comp}}) \cdot f_h \end{aligned} \tag{58}$$

where f_h is the harmonic order.

6 RMS Simulation

6.1 Zero sequence mutual impedance

In case of a zero sequence mutual impedance is defined (see chapter 4.1).

6.2 Consideration of Line Shunt Compensation

Consideration of line shunt compensation in RMS simulation is as described for the load flow calculation. Please refer to Section 3.5.

7 EMT Simulation

The models described in the previous sections are defined in the frequency domain and used in *PowerFactory* for all steady-state calculations such as load flow, short-circuit, harmonic load flow, frequency sweep, and the electromechanical (RMS) simulation.

This section introduces the models used in the electromagnetic transients (EMT) simulation. These models are obtained via the conversion of frequency-domain models into time-domain models.

7.1 Lumped Parameter Model

The lumped parameter model discussed in Section 2.1 can be directly used in EMT simulations by replacing the $j\omega$ operator by d/dt and thus the impedances and admittances by the corresponding inductances and susceptances.

For the pi-equivalent circuit in Figure 2.1, equations (5), (6) and (7) then become:

$$\begin{bmatrix} u_{s,A} \\ u_{s,B} \\ u_{s,C} \end{bmatrix} - \begin{bmatrix} u_{r,A} \\ u_{r,B} \\ u_{r,C} \end{bmatrix} = \begin{bmatrix} \Delta u_A \\ \Delta u_B \\ \Delta u_C \end{bmatrix} = \begin{bmatrix} R_s & R_m & R_m \\ R_m & R_s & R_m \\ R_m & R_m & R_s \end{bmatrix} \cdot \begin{bmatrix} i_A \\ i_B \\ i_C \end{bmatrix} + \begin{bmatrix} L_s & L_m & L_m \\ L_m & L_s & L_m \\ L_m & L_m & L_s \end{bmatrix} \cdot \frac{d}{dt} \begin{bmatrix} i_A \\ i_B \\ i_C \end{bmatrix} + \begin{bmatrix} u0 \\ u0 \\ u0 \end{bmatrix} \tag{59}$$

$$\begin{bmatrix} \Delta i_{s,A} \\ \Delta i_{s,B} \\ \Delta i_{s,C} \end{bmatrix} = \frac{1}{2} \left(\begin{bmatrix} G_s & G_m & G_m \\ G_m & G_s & G_m \\ G_m & G_m & G_s \end{bmatrix} \cdot \begin{bmatrix} u_{s,A} \\ u_{s,B} \\ u_{s,C} \end{bmatrix} + \begin{bmatrix} C_s & C_m & C_m \\ C_m & C_s & C_m \\ C_m & C_m & C_s \end{bmatrix} \cdot \frac{d}{dt} \begin{bmatrix} u_{s,A} \\ u_{s,B} \\ u_{s,C} \end{bmatrix} \right) \tag{60}$$

$$\begin{bmatrix} \Delta i_{r,A} \\ \Delta i_{r,B} \\ \Delta i_{r,C} \end{bmatrix} = \frac{1}{2} \left(\begin{bmatrix} G_s & G_m & G_m \\ G_m & G_s & G_m \\ G_m & G_m & G_s \end{bmatrix} \cdot \begin{bmatrix} u_{r,A} \\ u_{r,B} \\ u_{r,C} \end{bmatrix} + \begin{bmatrix} C_s & C_m & C_m \\ C_m & C_s & C_m \\ C_m & C_m & C_s \end{bmatrix} \cdot \frac{d}{dt} \begin{bmatrix} u_{r,A} \\ u_{r,B} \\ u_{r,C} \end{bmatrix} \right) \tag{61}$$

In case of a zero sequence mutual impedance are defined (see chapter 4.1). An addition zero sequence voltage $u0$ is added, if not $u0 = 0$.

Where

$$U0 = \sum_{i}^{n} (R0m_i \cdot I0_i + L0m_i \cdot \frac{d}{dt} I0_i) \quad kV$$

$$u0 = \frac{U0 \cdot \sqrt{3}}{\sqrt{2} \cdot Unom} \quad p.u.$$

and

- i index of coupled lines

- $R0m_i$ is the zero sequence mutual resistance in Ω
- $L0m_i$ is the zero sequence mutual inductance in H
- $I0_i$ is the zero current in kA of the coupled lines
- $Unom$ is the nominal voltage in kV of connected terminal i

The zero sequence current $I0_i$ is negated in case of the orientation is in opposite direction.

7.2 Distributed Parameter Model

The distributed parameter model cannot be directly used for EMT simulations because the elements of the equivalent circuit are a function of the frequency, as shown in equations (39). To make the model usable for EMT simulations, further assumptions have to be made. The resulting distributed parameters models are described below.

The EMT models of distributed parameter lines are based on Bergeron's method for solution in the time domain. The following options are supported:

- Constant parameter model
- Frequency dependent parameter (modal domain)
- Frequency dependent parameter (phase domain)

These options and their associated settings can be found on the *EMT-Simulation* page of the line element (*ElmLne*) and the line coupling (*ElmTow*) element. The frequency dependent parameter (phase domain) model is available for three-phase, single-phase, AC/DC and DC lines.

7.2.1 Bergeron's Method for Solution in the Time Domain

Considering the border conditions depicted in Figure 2.13, (35) can be written as:

$$U_r = \left(\frac{U_s - I_s \cdot Z_C}{2}\right) \cdot e^{\gamma \cdot l} + \left(\frac{U_s - I_s \cdot Z_C}{2}\right) \cdot e^{-\gamma \cdot l} \tag{62}$$

$$Z_C \cdot I_r = \left(\frac{U_s - I_s \cdot Z_C}{2}\right) \cdot e^{\gamma \cdot l} - \left(\frac{U_s + I_s \cdot Z_C}{2}\right) \cdot e^{-\gamma \cdot l} \tag{63}$$

and subtracting 63 from 62:

$$U_r - Z_C \cdot I_r = (U_s + Z_C \cdot I_s) \cdot e^{-\gamma \cdot l} \tag{64}$$

or rewritten as

$$I_r = \frac{U_r}{Z_C} - \left(I_s + \frac{U_s}{Z_C}\right) \cdot e^{-\gamma \cdot l} \tag{65}$$

The expression $U + Zc \cdot I$ of the border condition at the sending end, *s*, is the same at the receiving end, *r*, after multiplication with the propagation factor $e^{-\gamma \cdot l}$.

By repeating this procedure and setting the initial conditions at node r, and then travelling with the wave from node r to node s, we obtain:

$$U_s - Z_C \cdot I_s = (U_r + Z_C \cdot I_r) \cdot e^{-\gamma \cdot l} \tag{66}$$

or rewritten as

$$I_s = \frac{U_s}{Z_C} - \left(I_r + \frac{U_r}{Z_C} \right) \cdot e^{-\gamma \cdot l} \tag{67}$$

Equations (64) and (66), or equations (65) and (67), define Bergeron's equations in the frequency domain. The method models the line using controlled current sources, *J*, with parallel admittances, Y_C, at both ends as shown in Figure 7.1 or alternatively, using a controlled voltage source, *V*, in series with the impedance, Z_C, as shown in Figure 7.2, where:

$$J_r = \left(I_r + \frac{U_r}{Z_C} \right) \cdot e^{-\gamma l} \tag{68}$$

$$V_r = (U_r + I_r \cdot Z_C) \cdot e^{-\gamma l} \tag{69}$$

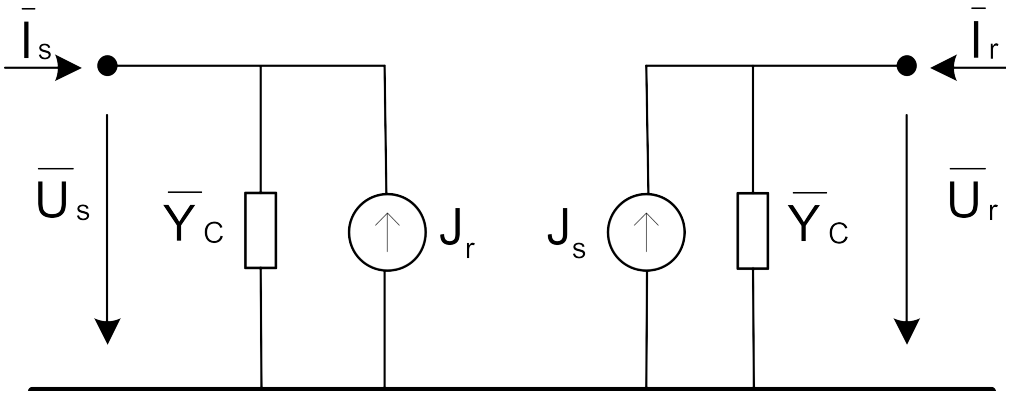


Figure 7.1: Bergeron's method: equivalent circuit with controlled current sources

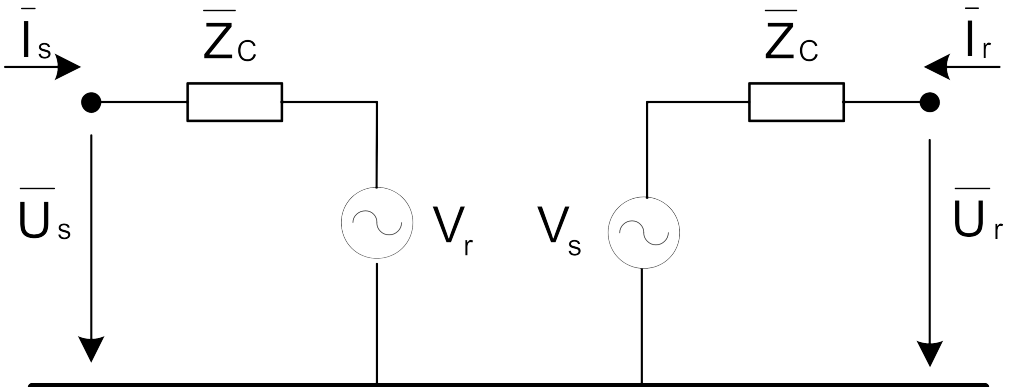


Figure 7.2: Bergeron's method: equivalent circuit with controlled voltage sources

The inverse Fourier transform is applied to transform the set of equations into the time domain:

$$u_s(t) = F^{-1} \left\{ Z_C \cdot I_s + (U_r + Z_C \cdot I_r) \cdot e^{-\gamma \cdot l} \right\} \tag{70}$$

$$u_r(t) = F^{-1}\left\{Z_C \cdot I_r + (U_s + Z_C \cdot I_s) \cdot e^{-\gamma \cdot l}\right\} \tag{71}$$

or rewritten:

$$i_s(t) = F^{-1}\left\{\frac{U_s}{Zc} - \left(I_r + \frac{U_r}{Zc}\right) \cdot e^{-\gamma \cdot l}\right\} \tag{72}$$

$$i_r(t) = F^{-1}\left\{\frac{U_r}{Zc} - \left(I_s + \frac{U_s}{Zc}\right) \cdot e^{-\gamma \cdot l}\right\} \tag{73}$$

where both the characteristic impedance:

$$Z_C = Z_C(\omega) = \sqrt{\frac{R' + j\omega \cdot L'}{G' + j\omega \cdot C'}} \tag{74}$$

and the propagation constant:

$$\gamma = \gamma(\omega) = \alpha(\omega) + j\beta(\omega) = \sqrt{(R' + j\omega \cdot L') \cdot (G' + j\omega \cdot C')} \tag{75}$$

are frequency dependent, even for constant per unit length line parameters R', L', G' and C'.

7.2.2 Constant Parameter Model

The constant distributed parameter model in *PowerFactory* is based on Bergeron's method, which calculates the voltages and currents at one end of the line based on the voltage and current at the other end delayed in time (i.e. the history term). Specifically, *PowerFactory* uses the lossy constant distributed parameter model described in [3]. This model is valid for lossy lines with a series resistance, R', and a negligible shunt conductance (a defined shunt conductance is considered internally as lumped at each terminal but not explicitely as part of the constant parameter model described below). These kinds of lines can then be modelled as single or multiple sections of lossless lines with resistances lumped in three places: $R/4$ at both ends of the line, and $R/2$ in the middle [3]. This model provides sufficient accuracy only if $R/4 << Z$, and hence should not be used for lines with very high resistance. *PowerFactory* emits an error when $R/4 > Z$, and a warning when $R/4 > 0.05 \cdot Z$.

The surge impedance is no longer calculated according to (74), but instead as:

$$Z_C = \sqrt{\frac{L'}{C'}} \tag{76}$$

being real and constant. The damping coefficient $\alpha = 0$, and hence from (75):

$$\gamma = j\beta = j\omega\sqrt{L' \cdot C'} \tag{77}$$

The propagation velocity is the same regardless of frequency and is given by:

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{L' \cdot C'}} \tag{78}$$

which allows the following definition of a travel time (frequency independent):

$$\tau = \frac{l}{v} = l \cdot \sqrt{L' \cdot C'} \tag{79}$$

In terms of travel time, the propagation constant can be rewritten as:

$$\gamma \cdot l = j\beta \cdot l = j\omega\sqrt{L' \cdot C'} \cdot l = j\omega \cdot \tau \tag{80}$$

and after simplifications we can rewrite (70) and (71) as:

$$\begin{aligned} U_s &= Z_C \cdot I_s + V_r \\ U_r &= Z_C \cdot I_r + V_s \end{aligned} \tag{81}$$

with:

$$\begin{aligned} V_r &= (U_r + Z_C \cdot I_r) \cdot e^{-j\omega\cdot\tau} \\ V_s &= (U_s + Z_C \cdot I_s) \cdot e^{-j\omega\cdot\tau} \end{aligned} \tag{82}$$

The inverse Fourier transform of the phase shift $e^{-j\omega\cdot\tau}$ in the frequency domain becomes a delay τ in the time domain, and the set of equations (81) and (82) transforms to:

$$\begin{aligned} u_s(t) &= Z_C \cdot i_s(t) + u_s(t-\tau) \\ u_r(t) &= Z_C \cdot i_r(t) + u_r(t-\tau) \end{aligned} \tag{83}$$

where the history terms are defined as:

$$\begin{aligned} u_s(t-\tau) &= u_r(t-\tau) + Z_C \cdot i_r(t-\tau) \\ u_r(t-\tau) &= u_s(t-\tau) + Z_C \cdot i_s(t-\tau) \end{aligned} \tag{84}$$

This can be reformulated in terms of current sources:

$$\begin{aligned} i_s(t) &= \tfrac{u_s(t)}{Z_C} + i_s(t-\tau) \\ i_r(t) &= \tfrac{u_r(t)}{Z_C} + i_r(t-\tau) \end{aligned} \tag{85}$$

where the history terms are defined as:

$$\begin{aligned} i_s(t-\tau) &= -\tfrac{u_r(t-\tau)}{Z_C} - i_r(t-\tau) \\ i_r(t-\tau) &= -\tfrac{u_s(t-\tau)}{Z_C} - i_s(t-\tau) \end{aligned} \tag{86}$$

The lumping of resistances according to the lossy model requires the modification of the impedance according to [3]:

$$Z_{mod} = Z + \frac{R}{4} \tag{87}$$

where Z is calculated according to (76). This results in the following modifications to the history terms in (86) [3]:

$$\begin{aligned} i_s(t-\tau) &= -\tfrac{Z}{Z_{mod}^2}[u_r(t-\tau) + (Z - \tfrac{R}{4}) \cdot i_r(t-\tau)] - \tfrac{R/4}{Z_{mod}^2}[u_s(t-\tau) + (Z - \tfrac{R}{4}) \cdot i_s(t-\tau)] \\ i_r(t-\tau) &= -\tfrac{Z}{Z_{mod}^2}[u_s(t-\tau) + (Z - \tfrac{R}{4}) \cdot i_s(t-\tau)] - \tfrac{R/4}{Z_{mod}^2}[u_r(t-\tau) + (Z - \tfrac{R}{4}) \cdot i_r(t-\tau)] \end{aligned} \tag{88}$$

In *PowerFactory*, the settings for this model can be adjusted on the *EMT-Simulation* page of the line element (*ElmLne*) or line coupling element (*ElmTow*) as follows:

- Line Model: Constant parameter
- Frequency for travel time estimation: enter a representative frequency for the transient under analysis. This frequency is used in (59) to calculate the propagation constant. In the case of a non-transposed line, the modal transformation matrix will also be calculated at this frequency.

***Note**: press the* Calculate Line Parameters *button any time you modify these parameters.* PowerFactory *will then calculate the propagation factor and the surge impedance at the specified frequency and initialise the model.*

7.2.3 Frequency Dependent Parameter (Modal) Model

With the exception of lossless and distortion-less lines, the characteristic impedance, Z_C, and propagation constant, γ, are frequency dependent. The variation of Zc and γ with frequency is most pronounced in the zero sequence mode and hence frequency-dependent models should be used when zero sequence currents or voltages are involved (i.e. in a single phase-to-ground fault).

Approximation by Rational Functions

To handle frequency-dependent parameters, *PowerFactory* uses the approach proposed by J. Marti [3, 7]. The characteristic impedance and the propagation factor are developed in rational functions and then the poles and zeros of the rational expressions calculated using a Bode approximation.

For the propagation factor, $A(\omega) = e^{-\gamma(\omega)\cdot l}$:

$$A_{app}(s) = e^{-s\cdot\tau_{\min}} \cdot k \cdot \frac{(s+z_1)\cdot(s+z_2)\ldots(s+z_n)}{(s+p_1)\cdot(s+p_2)\ldots(s+p_m)} \tag{89}$$

with $s = j\omega$ and $n < m$.

Similarly for the characteristic impedance, Z_C:

$$Z_{c-app}(s) = k \cdot \frac{(s+z_1)\cdot(s+z_2)\ldots(s+z_n)}{(s+p_1)\cdot(s+p_2)\ldots(s+p_n)} \tag{90}$$

with $s = j\omega$. This expression can then be expanded into partial fractions:

$$Z_{c-app}(s) = k_0 + \frac{k_1}{(s+p_1)} + \frac{k_2}{(s+p_2)} \cdots \frac{k_n}{(s+p_n)} \tag{91}$$

The accuracy of the model depends on the quality of the rational function approximations for A and Zc. To verify the approximation *PowerFactory* provides plots the exact and approximated solutions of A and Z_c on the *EMT-Simulation* page (*Mode* tabs) of the line (*ElmLne*) and line coupling (*ElmTow*) elements as shown in Figure 7.3. Note that the plots for each mode are displayed on their own tabbed pages. Right click on the plot to zoom in and out.



Figure 7.3: Bode approximations of A and Zc for the zero-sequence

Solution in the Time Domain

In this section, only equations for the equivalent circuit with current sources are described. Similar equations are used for the equivalent circuit with voltage sources.

Explicitly writing the frequency-dependent parameters, the input current at node *s* is:

$$I_s = \frac{U_s}{Z_C(\omega)} + J_r(\omega)$$

$$J_r(\omega) = -\left(I_r + \frac{U_r}{Z_C}\right) \cdot A(\omega)$$

The inverse Fourier transform of the controlled current source, J_r, can be evaluated by means of the convolution integral and hence:

$$j_r(t) = -\int_0^{\infty} \left[i_r(t-u) + \frac{i_r(t-u)}{Z_c}\right] \cdot a(u) \cdot du \tag{92}$$

with $a(t) = F^{-1}\{A(\omega)\}$, $\tau\min$ is the travel time of the fastest waves and $\tau\max$ is the travel time of the slowest waves. The convolution integral then only needs to be evaluated between $\tau\min$ and $\tau\max$ because $a(t)$ is zero up to $t = \tau\min$ and tends to zero for $t \to \tau\max$.

$a(t)$ is the inverse Fourier transform of $A(\omega)$. After $A(\omega)$ is expanded into partial fractions, the inverse Fourier transform then becomes a sum of exponentials:

$$a_{app}(t) = \begin{cases} 0 & \text{for } t < \tau_{\min} \\ k_1 e^{-p_1(t-\tau_{\min})} + k_2 e^{-p_2(t-\tau_{\min})} \ldots k_m e^{-p_m(t-\tau_{\min})} & \text{for } t \geq \tau_{\min} \end{cases} \tag{93}$$

Similarly, the inverse Fourier transform of (91) results in exponential terms of the form $e^{-t/RC}$ that corresponds to an RC network as shown in Figure 7.4 where:

$$R_0 = k_0 \;\; and \;\; R_i = \frac{k_i}{p_i}, \;\; C_i = \frac{1}{k_i} \;\; with \;\; i = 1 \ldots n \tag{94}$$

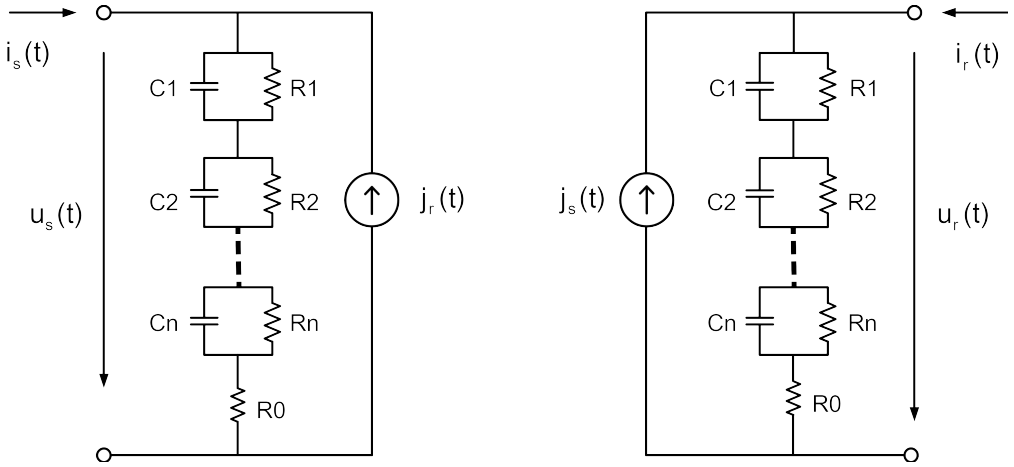


Figure 7.4: Equivalent circuit for the frequency-dependent parameter model

Then with $a(t)$ being a sum of exponential functions and Z_c developed as an RC-network, equation (92) can be solved using recursive convolution.

Data for the distributed frequency-dependent parameters model can be adjusted on the *EMT-Simulation* page of the line element (*ElmLne*) or line coupling element (*ElmTow*) as follows:

- Line Model: Frequency dependent parameter
- Frequency for travel time estimation: enter a representative frequency for the transient under analysis. The modal transformation matrix is calculated at this frequency, and afterwards remains real and constant.
- Min and Max. Frequency of parameter fitting: enter the minimum and maximum frequency for the approximation by rational functions of the propagation factor (82) and the characteristic impedance (85).
- Tolerance for Bode approximation: the maximum allowable error in % for the Bode approximation of the propagation factor (82) and the characteristic impedance (85). The lower the tolerance the higher the number poles and zeros of the approximated rational expressions.

***Note**: press the* Calculate Line Parameters *button any time you modify these parameters or enter new ones.* PowerFactory *will then calculate the propagation factor and the characteristic impedance at the specified frequency and set up the model.*

7.2.4 Frequency Dependent Phase-Domain Models

In general, transmission lines can be characterised by two frequency-dependent matrix transfer functions:

- The propagation function, $\mathbf{A}(\omega)$

- The characteristic admittance, $\mathbf{Y}_c(\omega)$ (or characteristic impedance $\mathbf{Z}_c(\omega)$)

These are expressed mathematically as:

$$\mathbf{A} = e^{-\sqrt{\mathbf{YZ}}\cdot l} \tag{95}$$

and

$$\mathbf{Y}_c = \mathbf{Z}^{-1} \cdot \sqrt{\mathbf{ZY}} \tag{96}$$

where $\mathbf{Z}$ and $\mathbf{Y}$ are the impedance and admittance matrices, respectively, and l is the line length. The time-domain simulation could be formulated using the inverse Fourier transform of the above functions and then solving the associated equations in the time domain using numerical convolution. However, the preferred time-domain solution uses rational function approximations of low order to ensure a computationally-efficient solution. In *PowerFactory*, these rational function approximations are obtained using vector fitting [5].

7.2.4.1 Rational Function Approximation

The accuracy of the rational function approximations of $\mathbf{A}$ and $\mathbf{Y}_c$ (or $\mathbf{Z}_c$) strongly influences the quality of the time-domain solution. The fitting of $\mathbf{A}$ and $\mathbf{Y}_c$ (or $\mathbf{Z}_c$) in the phase domain means that the phase-domain transfer functions will be intrinsically stable [4].

7.2.4.2 Approximation of $\mathbf{Y}_c$ or $\mathbf{Z}_c$

The characteristic admittance, $\mathbf{Y}_c$, (or characteristic impedance, $\mathbf{Z}_c$), are smooth functions of frequency, hence the fitting is straightforward and can be carried out directly in the phase domain [2]. The trace of $\mathbf{Y}_c$ is fitted in order to calculate the poles, and the residues are subsequently calculated per element (k,l) of $\mathbf{Y}_c$ using the following approximation:

$$(\mathbf{Y}_c(s))_{k,l} = d + \sum_{m=1}^{N} \frac{r_{i,m}}{s - p_{i,m}} \tag{97}$$

where constant d is real, N is the number of poles, $r_{i,m}$ are the residues, and $p_{i,m}$ are the poles. The residues and poles may be real or in complex conjugate pairs. Fitting is performed in the phase domain using vector fitting, and all elements of $\mathbf{Y}_c$ get identical poles. This procedure is used for the fitting of $\mathbf{Y}_c$ (or $\mathbf{Z}_c$) for both the ULM and FDM.

7.2.4.3 Universal Line Model

The Universal Line Model (ULM) proposed in [2] offers high accuracy and a phase-domain formulation.

Approximation of $\mathbf{A}$ The fitting of the propagation function, $\mathbf{A}$, is difficult because it contains modal components which have differing time delays [2]. In practicality, the differences in the time delays can be significant in the case of cables. This is due to the different permittivities in the insulation. Hence, the approximation of $\mathbf{A}$ is performed using a two-step approach; (i) fitting in the modal domain; followed by (ii) final fitting in the phase domain.

Modal-Domain Fitting of $\mathbf{A}$: The modes of $\mathbf{A}$ are calculated using a frequency-dependent transformation matrix, $\mathbf{T}$. The modal-domain fitting is comprised of three procedures:

Backwinding: Multiplication with a factor, $e^{(j\cdot\omega\cdot\tau)}$, in order to remove most of the oscillatory behaviour of the elements of $\mathbf{A}$. These elements are oscillating functions of frequency due to the time delay of the line [4]. The diagonal elements of the modal propagation matrix, $\mathbf{A}^m$, can be expressed as [8]:

$$a_i^m(\omega) = e^{-\alpha_i(\omega)+j\cdot\frac{\omega}{v_i(\omega)}\cdot l} \tag{98}$$

where α is the attenuation, v is the velocity, j is the imaginary unit and l is the line length. Each mode i is approximated by:

$$a_i^m(s) \approx \sum_{m=1}^{N} \frac{r_{i,m}}{s-p_{i,m}} \cdot e^{-s\cdot\tau_i} \tag{99}$$

where N is the number of poles for mode i, $r_{i,m}$ are the residues, and $p_{i,m}$ are the poles. The residues and poles may be real or in complex conjugate pairs.

Collapsing: The process of replacing modes having almost equal time delays with a single mode equal to the average of the modes. The criterion used for the formation of groups is provided in [2]. The concept of "groups" thereby replaces that of modes.

Fitting: Vector fitting is used to obtain the rational function approximation and all poles contribute to all elements of $\mathbf{A}$.

Phase-Domain Fitting of $\mathbf{A}$: The final fitting of $\mathbf{A}$ (i.e. the calculation of residues) is done in the phase domain using the poles and time delays found via fitting in the modal domain. Each element has the form:

$$\mathbf{A}(s) \approx \sum_{i=1}^{n}\left(\sum_{m=1}^{N} \frac{r_{i,m}}{s-p_{i,m}}\right) \cdot e^{-s\cdot\tau_i} \tag{100}$$

where n is now the number of groups, and the poles, $p_{i,m}$, and the time delays, τ_i are known.

7.2.4.4 Frequency Dependent Model (FDM)

The Frequency Dependent Model (FDM) introduced in [6] offers improved accuracy and stability via the simultaneous fitting of poles and residues in the phase domain. High residue-pole ratios are thereby avoided, making the FDM less prone to divergence due to such high ratios in EMT simulations.

Approximation of $\mathbf{A}$ The difficulties associated with the fitting of the propagation function, $\mathbf{A}$, were mentioned in Section 7.2.4.3. The approximation of $\mathbf{A}$ for the FDM is performed by fitting the modal contributions in the phase domain using a common set of poles for each modal contribution, i, below. The poles and residues of $\mathbf{A}$ are obtained simultaneously, as follows:

$$\mathbf{A}(s) \cong \sum_{i=1}^{n} \hat{\mathbf{A}}_i \cdot e^{-s\cdot\tau_i} \tag{101}$$

where n is the number of modal contribution groups. Fitting is then performed with the constant time delay, τ_i, removed:

$$\hat{\mathbf{A}}_i(s) \approx \sum_{j=1}^{M_i} \frac{\mathbf{R}_{i,j}}{s-p_{i,j}} \tag{102}$$

where M_i is the order of the approximation for the ith modal propagation function.

Fitting Procedure The following steps are repeated, increasing the number of poles until either: (i) the user-defined RMS error threshold has been met; or (ii) the user-defined maximum number of poles has been reached:

1. Vector fitting is applied iteratively to reduce the RMS error. Iteration stops if either the user-defined maximum RMS error or the user-defined maximum number of iterations is reached.
2. If using option *Increase and flip unstable poles*, unstable poles are flipped to the left-half plane and the RMS error is recalculated based on the resulting set of poles. This generally worsens the RMS error, as expected.
3. If using option *Increase until rms error is met*, the RMS error is calculated based on the unadjusted set of poles. This generally results in a lower RMS error than the previous option, however sometimes at the cost of unstable poles which can result in time-domain divergence in some cases.
4. For both of these options, if the RMS error does not meet the RMS error threshold, the number of poles is increased, and the algorithm continues from the first step.

Following the fitting process, *PowerFactory* reports the RMS error and the poles obtained by vector fitting (if option *Output poles* is selected).

The stability of the time-domain solution is dependent not only upon the accuracy of the rational function approximation (i.e. the resulting RMS error), but also upon the proximity of the poles to each other, and the ratio between residue and pole.

For $\mathbf{Y}_c$ and $\mathbf{Z}_c$, a worse RMS error is usually obtained when fitting starting at very low frequencies, and a better RMS error is usually obtained using a higher-order approximation (i.e. more poles).

For $\mathbf{A}$, a low number of poles is usually sufficient for accurate time-domain simulation, which assists in avoiding high residue-to-pole ratios that tend to occur with higher-order approximations (in particular for the ULM).

7.2.4.5 Options

The following inputs are available for vector fitting:

- **Min. Frequency of Parameter Fitting**: Lower bound of the frequency range for the fitting in Hz
- **Max. Frequency of Parameter Fitting**: Upper bound of the frequency range for the fitting in Hz
- **RMS error**: The target RMS error to be obtained by the vector fitting.
- **Max. number of iterations**: The maximum number of iterations used when performing the vector fitting.
- **Max. number of poles**: The maximum number of poles allowed. A larger number of poles may increase accuracy but will cause a subsequently slower performance.
- **Number of poles**: The number of poles found will be determined by which option is selected:
  - Increase and flip unstable poles: The number of poles will be increased and if any pole resides in the right half (unstable region) of the complex plane it will be flipped into the left-half plane.

  - Increase until rms met: The number of poles will be increased until the target rms error is met. This may yield a lower rms error than using *Increase and flip unstable poles* at the risk of finding poles in the right-half plane (i.e. unstable poles). These poles will likely cause time-domain divergence but can be seen if option *Output poles* (described below) is selected. Hence, this option must be used with caution.

- **Stability margin**: The margin from the vertical axis at which poles will be placed. For example, an entry of -1 will allow any negative pole to be placed no closer than this to the vertical (imaginary) axis in the left-half plane.
- **Fine-tune delays**: Applies an optimisation procedure to improve the vector fitting by readjusting the time delays together with the poles and residues.
- **Apply DC correction**: Improves the fit near DC via the use of a functional reformulation.
- **Output poles**: The poles will be printed in the output window.

7.3 Diagonalisation

The modal-domain models presented in §7.2 implicitly assume a single-phase line. In reality however, distributed parameter models are required for three-phase long lines or for transmission systems with multiple 3-phase circuits. To handle this, equations (59) - (61) have to be diagonalised. After diagonalisation, the mutually-coupled equations of the 3-phase system transform to 3 independent, and hence decoupled, single-phase systems.

The diagonalisation in *PowerFactory* is carried out as follows:

- Balanced lines: the impedance and admittance matrices of these lines are diagonal-cyclic, i.e. Z/Y-matrices of the form (8). This is normally the case for transposed lines. To diagonalise the matrices, *PowerFactory* uses the transformation into symmetrical components according to (19). The transformation matrix is known a-priori and is constant (not frequency dependent).
- Unbalanced lines: this is typically the case for untransposed lines. The Z/Y-matrices are no longer diagonal-cyclic; hence to diagonalise them, a transformation into modal components is required. In this case, the transformation matrices are not known a-priori but are determined from an eigenvalue and eigenvector calculation. In this case, the transformation matrices are frequency dependent.

For unbalanced lines and steady-state calculations (i.e. harmonic load flow or frequency sweep), *PowerFactory* calculates the transformation matrices, and therefore the eigenvalues and eigenvectors, at each frequency of interest to account for the frequency dependency of the transformation matrices.

In the EMT-Simulation, the transformation matrix is calculated at a single frequency, i.e. at the one specified by the user on the *EMT-Simulation* page of the *ElmLne* or *ElmTow* element dialog. The transformation matrix then remains constant. Furthermore the approximation of the complex transformation matrix (eigenvectors) is made only using the real part.

7.4 Consideration of Line Shunt Compensation

The following two configurations are considered:

1. The line compensation is a shunt reactor (i.e. when $Q1 > 0$)

2. The line compensation is a shunt capacitor

7.4.0.6 Shunt Reactor

The current through the reactor is first transformed to the $\alpha\beta 0$ system, and the voltage across the reactor is then calculated as follows:

$$\begin{aligned} u_{L_\alpha} &= \Re(\underline{z}_{1_{comp}}) \cdot i_\alpha + \Im(\underline{z}_{1_{comp}})/\omega_0 \cdot \frac{\mathrm{d}i_{L_\alpha}}{\mathrm{d}t} \\ u_{L_\beta} &= \Re(\underline{z}_{1_{comp}}) \cdot i_\beta + \Im(\underline{z}_{1_{comp}})/\omega_0 \cdot \frac{\mathrm{d}i_{L_\beta}}{\mathrm{d}t} \\ u_{L_0} &= \Re(\underline{z}_{0_{comp}}) \cdot i_0 + \Im(\underline{z}_{0_{comp}})/\omega_0 \cdot \frac{\mathrm{d}i_{L_0}}{\mathrm{d}t} \end{aligned} \tag{103}$$

The α, β, 0 voltages are then transformed back to phase quantities (u_{L_a}, u_{L_b}, u_{L_c}) to formulate the differential equations using u_a, u_b and u_c (the terminal voltages of the line end to which the reactor is connected). The corresponding impedances are obtained from the admittances:

$$\begin{aligned} \underline{z}_{1_{comp}} &= \frac{1}{\underline{y}_{1_{comp}}} \\ \underline{z}_{0_{comp}} &= \frac{1}{\underline{y}_{0_{comp}}} \end{aligned} \tag{104}$$

7.4.0.7 Shunt Capacitor

The currents though the capacitor are:

$$\begin{aligned} i_{C_\alpha} &= \Re(\underline{y}_{1_{comp}}) \cdot u_\alpha + \Im(\underline{y}_{1_{comp}})/\omega_0 \cdot \frac{\mathrm{d}u_\alpha}{\mathrm{d}t} \\ i_{C_\beta} &= \Re(\underline{y}_{1_{comp}}) \cdot u_\beta + \Im(\underline{y}_{1_{comp}})/\omega_0 \cdot \frac{\mathrm{d}u_\beta}{\mathrm{d}t} \\ i_{C_0} &= \Re(\underline{y}_{0_{comp}}) \cdot u_0 + \Im(\underline{y}_{0_{comp}})/\omega_0 \cdot \frac{\mathrm{d}u_0}{\mathrm{d}t} \end{aligned} \tag{105}$$

The voltages α, β, 0 are calculated from the terminal phase voltages (u_a, u_b, u_c) using a standard transformation. The currents through the capacitor are then transformed back to phase quantities and subtracted from the line currents, $i_{a,b,c}$.

A Parameter Definitions

Table A.1: Input parameter of the line type (*TypLne*)

Name	Description	Unit	Range	Default	Symbol
loc_name	Name				
uline	Rated Voltage	kV	x>=0	0	
sline	Rated Current	kA	x>0	1	
InomAir	Rated Current (in air)	kA	x>0	1	
frnom	Nominal Frequency	Hz	x>=0	50	
aohl_	Cable / OHL (overhead line)			cab	
systp	System Type		AC:DC	AC	
nlnph	No. of Phases		01:02:03	3	
nneutral	No. of Neutrals		00:01	0	
rline	Parameters per Length 1,2-Sequence: Resistance R' (20°C)	Ohm/km	x>=0	0	$R_1^{'}$
xline	Parameters per Length 1,2-Sequence: Reactance X'	Ohm/km		0	$X_1^{'}$
lline	Parameters per Length 1,2-Sequence: Inductance L'	mH/km	x>=0	0	$L_1^{'}$
rline0	Parameters per Length Zero Sequence: Resistance R0'	Ohm/km	x>=0	0	$R_0^{'}$
xline0	Parameters per Length Zero Sequence: Reactance X0'	Ohm/km	x>=0	0	$X_0^{'}$
lline0	Parameters per Length Zero Sequence: Inductance L0'	mH/km	x>=0	0	$L_0^{'}$
rnline	Parameters per Length, Neutral: Resistance Rn'	Ohm/km	x>=0	0	$R_n^{'}$
xnline	Parameters per Length, Neutral: Reactance Xn'	Ohm/km	x>=0	0	$X_n^{'}$
lnline	Parameters per Length, Neutral: Inductance Ln'	mH/km	x>=0	0	$L_n^{'}$
rpnline	Parameters per Length, Phase-Neutral Coupling: Resistance Rpn'	Ohm/km	x>=0	0	$R_{pn}^{'}$
xpnline	Parameters per Length, Phase-Neutral Coupling: Reactance Xpn'	Ohm/km	x>=0	0	$X_{pn}^{'}$
lpnline	Parameters per Length, Phase-Neutral Coupling: Inductance Lpn'	mH/km	x>=0	0	$L_{pn}^{'}$
tmax	Parameters per Length 1,2-Sequence: Max. Operational Temperature	°C	x>=20	80	
rline_tmax	Parameters per Length 1,2-Sequence: Resistance R' at max. operational temperature	Ohm/km	x>=0	0	

alpha	Parameters per Length 1,2-Sequence: Temperature Coefficient	1/K	x>=0	0.00403	α
mlei	Parameters per Length 1,2-Sequence: Conductor Material			Al	
bline	Parameters per Length 1,2-Sequence: Susceptance B'	uS/km		0	$B_1^{'}$
cline	Parameters per Length 1,2-Sequence: Capacitance C'	uF/km		0	$C_1^{'}$
tline	Parameters per Length 1,2-Sequence: Ins. Factor			0	$tg\delta_1$
gline	Parameters per Length 1,2-Sequence: Conductance G'	uS/km		0	$G_1^{'}$
bline0	Parameters per Length Zero Sequence: Susceptance B0'	uS/km		0	$B_0^{'}$
cline0	Parameters per Length Zero Sequence: Capacitance C0'	uF/km		0	$C_0^{'}$
Ices	Parameters per Length Zero Sequence: Earth-Fault Current	A/km		0	
tline0	Parameters per Length Zero Sequence: Ins. Factor			0	$tg\delta_0$
gline0	Parameters per Length Zero Sequence: Conductance G0'	uS/km		0	$G_0^{'}$
bnline	Parameters per Length, Neutral: Susceptance Bn'	uS/km	x>=0	0	$B_n^{'}$
cnline	Parameters per Length, Neutral: Capacitance Cn'	uF/km	x>=0	0	$C_n^{'}$
bpnline	Parameters per Length, Phase-Neutral Coupling: Susceptance Bpn'	uS/km	x>=0	0	$B_{pn}^{'}$
cpnline	Parameters per Length, Phase-Neutral Coupling: Capacitance Cpn'	uF/km	x>=0	0	$C_{pn}^{'}$
rtemp	Max. End Temperature	°C	x>0	80	
Ithr	Rated Short-Time (1s) Current (Conductor)	kA	x>=0	0	
picln	Inrush Peak Current: Ratio Ip/In	p.u.		0	
pitln	Inrush Peak Current: Maximum Time	s		0	
fcharC1	Frequency Dependency of Pos.-Sequence Capacitance: C1'(f) (ChaPol,ChaVec,ChaMat)				
fcharC0	Frequency Dependency of Zero-Sequence Capacitance: C0'(f) (ChaPol,ChaVec,ChaMat)				
fcharR1	Frequency Dependencies of Pos.-Sequence Impedance: R1'(f) (ChaPol,ChaVec,ChaMat)				

fcharL1	Frequency Dependencies of Pos.-Sequence Impedance: L1'(f) (ChaPol,ChaVec,ChaMat)				
fcharR0	Frequency Dependencies of Zero-Sequence Impedance: R0'(f) (ChaPol,ChaVec,ChaMat)				
fcharL0	Frequency Dependencies of Zero-Sequence Impedance: L0'(f) (ChaPol,ChaVec,ChaMat)				
pStoch	Stochastic model (StoTyplne)				
manuf	Manufacturer				
chr_name	Characteristic Name				
for_name	Foreign Key				
dat_src	Data source			MAN	
doc_id	Additional Data ()				
desc	Description				
miso	Insulation Material				
iopt_cnd	Cable is			mlt	
cmeth	Installation Method (IEC 364)			C	
iopt_ord	Conductors			tre	
qurs	Nominal Cross Section	mm*2		0	
cabdiam	Outer Diameter	mm		0	
iopt_dir	Arrangement			hor	
lcost	Line Cost	$/km	x>=0	0	

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List of Figures

1.1 Positive sequence input parameters . . . 3
2.1 Equivalent PI-circuit of the line for lumped parameters . . . 5
2.2 Lumped parameter model for a single-phase AC line . . . 6
2.3 Equivalent circuit of the single-phase line with neutral conductor model . . . 7
2.4 Equivalent circuit of the two-phase line model . . . 8
2.5 Equivalent circuit of the two-phase line with neutral conductor model . . . 9
2.6 Equivalent circuit of the three-phase line . . . 10
2.7 Equivalent circuit for the 3-phase line with neutral conductor . . . 11
2.8 Phase to phase measurement loop . . . 12
2.9 Neutral to ground measurement loop . . . 13
2.10 Phase – PE (earth wire) measurement . . . 13
2.11 Phase – neutral measurement . . . 14
2.12 Incremental model for a line of elemental length . . . 16
2.13 Equivalent pi-circuit for the line with distributed parameters in the frequency domain 17
4.1 Example for zero sequence mutual impedance coupling . . . 25
4.2 Orientation of zero sequence mutual impedance . . . 26
4.3 Short-circuit at line with zero sequence mutual impedance . . . 27
7.1 Bergeron's method: equivalent circuit with controlled current sources . . . 32
7.2 Bergeron's method: equivalent circuit with controlled voltage sources . . . 32
7.3 Bode approximations of A and Zc for the zero-sequence . . . 36
7.4 Equivalent circuit for the frequency-dependent parameter model . . . 37

List of Tables

1.1 Overview of line models available in *PowerFactory* 1

A.1 Input parameter of the line type (*TypLne*) 43